The Validation Bottleneck: Alpha Discovery When Hypotheses Are Free

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Abstract

Large language models have made investment hypotheses effectively free, moving the binding constraint on alpha discovery from generation to validation. Restricting a model's data feed does not restrict its knowledge: it has read the anomaly literature, so its training cutoff is the only valid out-of-sample boundary. Put the two facts together and the strategies tested multiply without limit while the evidence that can judge them accrues at one year per year. Three consequences follow. The *smallest* alpha an allocator can certify rises with the logarithm of the number of strategies the industry tests, because the luckiest fluke among them looks better the more are tried. Alpha that decays faster than the evidence accrues cannot be certified within its own lifetime. And firms drawing hypotheses from the same model correlate their discoveries, on that model's most celebrated ideas, before either holds a position. Two pre-registered experiments then measure the contamination and the bind. Told to propose predictors *not yet in the literature* as of a past date, five models across three lineages return that literature anyway: 26–55% of their "novel" proposals match predictors already in the library. Convergent rediscovery could explain that; it cannot explain the sign. The predictors they propose most often are those that performed *worse* afterward (combined $z = -3.88$). They reproduce fame, not future returns. Evaluated strictly after verified cutoffs, none of the roughly 900 evaluable strategies clears false-discovery control, and no model's best pick beats the best of an equal number of random draws from the same restricted language; two of the three do worse. The models' own stated predictions about their mechanisms held 39% of the time, against 70% for mechanisms already documented as true put through the same tests—and 46% against 86% on the one window long enough to test structure. By the second consequence, no window available today could certify a true strategy either: the nulls measure the window's width, not the models' quality. Hypotheses are free, and worth what they cost until validated.

Keywords: large language models, backtest overfitting, data contamination, multiple testing, market efficiency, factor investing.

JEL Classification: G11, G12, G14, C12, C58, O33.

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1 Introduction

For as long as quantitative investing has existed, its implicit production function has treated ideas as the scarce input. Hypotheses were costly—they required researchers, reading, and time—while the data on which hypotheses were tested was a common resource, drawn on freely by every participant. The institutions of the field reflect this arrangement. Journals referee the plausibility of mechanisms; firms hire researchers by their ability to originate signals; the multiple-testing literature polices the boundary where cheap variations on ideas meet a finite data supply [Harvey et al., 2016; Bailey and López de Prado, 2014; Harvey and Liu, 2021].

Large language models invert the arrangement. A frontier model, prompted appropriately, returns eight coherent factor hypotheses—each with a named construction, an economic mechanism, and testable implications—in under a minute, at a marginal cost measured in cents. The generation of investment hypotheses, as an economic activity, has collapsed to a price of approximately zero. This paper is about what that collapse does to the statistics and the economics of alpha discovery. Our thesis is that the binding constraint moves to the one input that no model, vendor, or budget can manufacture: *out-of-sample time*—observations that postdate the knowledge of whatever process generated the hypothesis. When hypotheses are free, validation capacity is the scarce factor, and the field reorganizes around it. We call this the *validation bottleneck*.

The argument proceeds in four steps. The first step is epistemic (Section 2). It is tempting to evaluate machine-generated strategies the way human-generated strategies have always been evaluated: on historical data. For an LLM this is invalid in principle, not merely biased in practice. An LLM trained through 2024 has read the anomaly literature in its entirety—every published predictor, every replication, every post-publication decay estimate. Restricting the model's *data feed* to 1963–1999 does not restrict its *knowledge* of what was discovered, published, and arbitraged away after 1999. Temporal segregation of data is not temporal segregation of knowledge. We state this as a contamination result and draw its operational corollary: the model's training cutoff is the only out-of-sample boundary that exists, and any evaluation period that predates the cutoff carries no evidential weight for the model's proposals. Moreover, by McLean and Pontiff [2016], the literature the model has absorbed documents predictability that substantially decayed upon publication. The model has, in a precise sense, read only the obituaries: its prior over "what works" is a prior over what *worked*, weighted by prominence rather than by surviving profitability. This observation yields a falsifiable prediction that our first experiment confirms.

The second step is statistical (Section 3). If genuine out-of-sample time begins only at the training cutoff, the researcher is left with a window a few years wide, and calendar time buys statistical power at a punishing rate. The relation is $t \approx \mathrm{SR}\sqrt{T}$. Put a number in it: a strategy whose true annualized Sharpe ratio is 0.5 needs roughly 36 years of forward data before its mean return clears $t > 3$. Nobody has 36 years. Waiting is not a research design, so the power has to come from somewhere other than the length of the window.

We develop two responses. The first is bookkeeping. Every time a generating process learns from validation outcomes—a prompt tuned toward what survived, a researcher who now knows which families pass—the window loses a little of its authority. That is the setting of *adaptive data analysis*, and the reusable-holdout theory of Dwork et al. [2015] supplies a principled budget: so many adaptive queries per window, then stop. To our knowledge that machinery has not previously been applied to financial research.

The second response is the one that does the real work. A mean return is the least of what a genuine mechanism implies. If the mechanism is real, the strategy should also behave differently across volatility regimes, fade in mega-caps or strengthen in small ones, correlate with named factors only up to a bound, and show up in both halves of the window. Each of those is a separate test, and—this is the point—all of them run *simultaneously* on the same short window. Under independence, k such auxiliary implications tested at sizes $\alpha_1, \ldots, \alpha_k$ leave a false mechanism a survival probability of at most $\prod_i \alpha_i$. Power accumulates in the cross-section of implications rather than in the time series of returns. This is, we argue, the only affordable source of statistical power in the bottleneck, and it is why an anti-HARKing protocol for machine-scale research must *require* auxiliary predictions rather than merely permit them.

A companion subsection lays out the statistical geography underneath all of this, because a widely held intuition about it is wrong. Selection inflation does not depend on breadth: the maximum backtest Sharpe ratio among M null strategies grows as $\sqrt{2 \ln M/T}$, to leading order, whether one times a single index or ranks three thousand stocks. Trading a wide cross-section is genuinely safer than timing one series, but not for the reason usually given. Two of the real defenses work by changing the terms of that expression—capping each candidate's complexity holds down the *effective* number of trials, and differencing out common paths preserves the effective sample length—while two more add tests the expression does not contain, namely internal replication and inspection of micro-parameters. The protocol below engineers all four deliberately.

The third step is economic (Section 4). In the equilibrium of Grossman and Stiglitz [1980], prices leave behind exactly enough predictability to compensate costly information acquisition. When acquisition—hypothesis generation—becomes free, that compensation has to attach to something else. We argue it attaches to *validation capacity*, and that a familiar statistical threshold becomes, read this way, an economic one. With M hypotheses tested industry-wide, the best pure fluke reaches a t-statistic of about $\sqrt{2 \ln M}$. An allocator cannot fund anything she cannot distinguish from that fluke. So the smallest fundable alpha rises with $\ln M$ and falls with the validation time available—a *certification floor*. The floor is not where prices settle; it is where the funding decision settles.

The underlying inequality is not ours. It is the minimum-backtest-length result of Bailey et al. [2014], and it appears as the exploitability threshold in Da et al. [2024], who also read statistical limits as equilibrium compensation. What we contribute is a reinterpretation of every symbol in it: M becomes the endogenous industry-wide trial count, inflated at machine speed; T becomes scarce *forward* time rather than an ever-growing historical sample; and the resulting threshold becomes the settling point of the funding process rather than a referee's hurdle.

Two further results follow. The first is a *knowability constraint*, and it concerns speed rather than width. Alpha decays while validation accumulates, and the two clocks compete. If a strategy's edge decays with half-life h from an initial Sharpe ratio SR_0, the largest t-statistic its sample mean can ever reach is about $0.77\,\mathrm{SR}_0\sqrt{h}$, no matter how long one waits. Certification at threshold t^* therefore requires $h \gtrsim 1.7\,(t^*/\mathrm{SR}_0)^2$ years. Some opportunities are thus unknowable in real time: they are gone before the evidence that they existed can be assembled, and validation quietly selects the slow decayers.

The second is a *generator monoculture* channel. When many firms draw hypotheses from the same handful of foundation models, correlation enters upstream of any position. The financial-stability literature locates such correlation in trading behavior [Kleinberg and Raghavan, 2021; European Central Bank, 2024; Dou et al., 2025; Meng and Chen, 2026b]; we locate it in discovery. Each firm's nominal count of trials explodes while the industry's effective count of *independent* trials collapses toward the shared generator's favorite ideas.

The fourth step is empirical (Section 5). Two experiments do the work, and both were pre-registered by design: generation corpora, evaluation walls, and analysis specifications were committed, in that order, before any performance data was examined. Every raw model response is archived.

Experiment 1 measures the contamination directly, by asking models to pretend it is an earlier date. Five models spanning three independent training lineages—OpenAI, Meta, and

Anthropic—were prompted to roleplay five past dates (January 1990 through January 2010) and propose predictors "not yet documented in the academic literature" as of each date. The resulting 2,750 proposals were matched against the 212-predictor open-source anomaly library of Chen and Zimmermann [2022]. The models cannot leave the literature they have read: between 26 and 55 percent of each model's "novel" proposals match a predictor already documented in the library, and among matches at the level of construction, 71–92 percent reproduce predictors published *after* the roleplay date. The sharper test asks a question no truthful roleplayer could answer: does the frequency with which a predictor is proposed line up with how that predictor *performed afterward*? It does, with the wrong sign. The relation is negative—rank correlation negative in 21 of 25 model-vintage cells, Fisher-combined $z = -3.88$. The models do not anticipate returns. They reproduce prominence.

Experiment 2 is the constructive counterpart: the only design the contamination result permits, executed. It begins by measuring, rather than trusting, each model's training cutoff; a recall probe finds confident knowledge extending up to five months past the reported dates. Four models then generated 1,336 composite factors inside a small, enumerable grammar built over the same public library, which makes the number of distinct trials exact rather than estimated ($M = 330$ for the longest-window model). Three of the four have walls early enough to be scored against existing data, contributing the 1,000 strategies actually evaluated below; the fourth is registered and waits for the library's next release.

The results are uniformly negative, and the negativity is informative. Evaluated strictly after the verified cutoffs, nothing clears dependence-robust false-discovery control at $q = 0.05$. Against 10,000 strategies drawn uniformly at random from each model's own grammar on the identical window, the best machine-selected strategy lands at the 9th, 58th, and 16th percentile of the best-of-M distribution. No generator improves detectably on random sampling from its own vocabulary: two land well below the null's midpoint and the third sits essentially on it. The models' mechanism-implied auxiliary predictions hold at 46 percent, where mechanisms documented as true in the published literature, run through the identical battery, pass at 70 percent. On the one window where evidence is possible, machine-scale generation produced no selection skill we can distinguish from chance. And the correct reading of that is not that the models are poor: it is that a window this narrow cannot certify anything at all, for models or for humans. That is the bottleneck binding.

Section 6 draws the institutional conclusion. The data commons on which all of this testing operates is depletable, and adaptive machine-scale querying depletes it at industry scale. We therefore describe a cross-firm testing registry whose unit of account is the reusable-holdout query budget. It upgrades the trial-disclosure proposals of López de Prado [2019a] and the registered-reports proposal of Harvey [2017] from norms of disclosure into a mechanism of governance.

Related literature. Four strands intersect here. We take them in turn and state precisely what is inherited from each.

Contamination. A fast-growing literature documents lookahead bias in LLM-based prediction. Sarkar and Vafa [2024] define the problem and demonstrate it with unpredictable-event tests; Glasserman and Lin [2024] decompose in-window bias in sentiment backtests; Lopez-Lira et al. [2025] show that in-sample, "any observed output is consistent with both genuine skill and memorization"; Gao et al. [2025b] supply a portable in-sample diagnostic; Li et al. [2025] document agent backtest returns evaporating past the knowledge window. All of this concerns the evaluation of *given* forecasts. None of it treats hypothesis *generation*, where contamination enters through the literature the model has read rather than through recall of particular outcomes, and none develops the economics of the resulting scarcity. The closest statement we have found is a single sentence in Gao et al. [2025b], noting that post-cutoff evaluation "limits statistical power."

Mitigation. A second strand tries to engineer the bias away: by subtracting recallable outcome memory at inference time [Li et al., 2026], by approximate unlearning over an enumerable forget-set [Merchant and Levy, 2025], or by retraining chronologically clean models from scratch [He et al., 2025; Drinkall et al., 2024; Yan et al., 2026]. We engage all three directly in Section 2. The first two operate on outcome-indexed memory and cannot reach selection on absorbed literature—Li et al. [2026] concede there is "no clean decomposition between memorised and reasoned components." Clean retraining does restore the exclusion restriction, but currently trades away the very capability that makes generation valuable.

Multiple testing and overfitting. This strand supplies our statistical raw material [Harvey et al., 2016; Bailey and López de Prado, 2014; Bailey et al., 2014, 2017; López de Prado and Bailey, 2018; López de Prado, 2019a]. The $\sqrt{2 \ln M}$ hurdle and the $2 \ln M/T$ length requirement are theirs. Our use of them is a reinterpretation, not a rederivation, and we mark the boundary explicitly wherever it falls.

Equilibrium. The nearest work in spirit prices information in equilibrium. Martin and Nagel [2022] elevate out-of-sample tests epistemically in a high-dimensional learning economy. Dugast and Foucault [2025] price costly *search* for predictors in a Grossman–Stiglitz equilibrium—precisely the margin that generation-at-zero-cost eliminates. Da et al. [2024] derive equilibrium bounds from statistical learning limits on a fixed historical panel. None of them makes forward validation time the scarce input, none models the generation stage, and none treats the knowability constraint; those are the margins claimed here.

Two recent papers are close complements rather than antecedents. Novy-Marx and Velikov [2026] industrialize the HARKing critique, with LLMs writing persuasive theory around data-mined signals, and Si et al. [2025b,a] document mode collapse and the ideation–execution gap in LLM research ideas. They establish that hypotheses are free and homogeneous. We establish what that makes scarce.

2 The Contamination Result and the Cutoff Wall

This section makes one claim and draws one consequence. The claim: a language model that has read the anomaly literature cannot be tested on the history that literature describes, no matter how carefully its inputs are restricted. The consequence, which we call the *cutoff wall*: the model's training cutoff is the only out-of-sample boundary that exists, so every backtest before that date carries zero evidential weight for the model's proposals. Everything else in the paper is a response to that one lost degree of freedom.

2.1 Temporal data segregation is not knowledge segregation

The idea in plain terms: you can hide data from a model, but you cannot hide what it already knows. Feed a model only pre-1999 prices and it still remembers that momentum crashed in 2009, that the accruals anomaly faded after Sloan [1996] became famous, and that low-volatility strategies were crowded once they were celebrated. Its hypotheses are selected in the light of that memory whether or not a single post-1999 number reaches its context window. The rest of this subsection states the point formally and names what it rules out.

Consider a researcher evaluating a hypothesis-generating process G—human or machine—by the familiar protocol: form hypotheses, construct the implied strategies, and measure their performance on a historical sample $\mathcal{D}_{[t_0,t_1]}$. The protocol's validity rests on an exclusion restriction: the generation of the hypothesis must not have used information from the evaluation period. For human researchers this restriction is policed, imperfectly, by careers and norms; the multiple-testing literature exists because it fails at the level of the profession [Harvey et al., 2016]. For an LLM the restriction fails at the level of the individual hypothesis, by construction.

Let $\mathcal{I}_G$ denote the information set of the generator and let τ_G denote the end of its training corpus—the training cutoff. For a pretrained language model, $\mathcal{I}_G$ includes, in compressed form, the contents of its corpus: the empirical finance literature through τ_G, the financial press through τ_G, and extensive summaries of realized market history through τ_G. Two properties of $\mathcal{I}_G$ drive everything that follows.

First, $\mathcal{I}_G$ cannot be truncated by prompt. Instructing the model to "use only information available in 1999," or feeding it only pre-1999 data, restricts its *inputs*, not its *parameters*. The knowledge that value crashed in the late 2010s, that accruals decayed after Sloan [1996] was absorbed by practitioners, that low-risk anomalies were arbitraged as they became famous—all of this shapes which hypotheses the model finds plausible, whether or not any post-1999 number appears in its context window. The general-ML literature has established this directly: Gao et al. [2025a] show that prompting a model to adopt an earlier knowledge cutoff fails precisely when the excluded content is causally related to the query rather than directly asked for; Lopez-Lira et al. [2025] show that masking fails because models reconstruct entities and dates from minimal context. Hypothesis selection is the extreme case of causally related knowledge: the model is not asked to recall any post-cutoff fact, yet every candidate it ranks is ranked *because of* the absorbed record of what followed.

Second, $\mathcal{I}_G$ is not enumerable from outside. One can audit a human research program's trials, at least in principle [López de Prado, 2019a]; one cannot enumerate the strategy evaluations implicit in a trillion-token corpus. The effective number of hypotheses "already tested" inside the generator is unobservable, which is fatal for any multiple-testing correction applied to a pre-cutoff backtest of its proposals.

These observations combine into a statement we will use as a theorem-in-practice.

Proposition 2.1 (Contamination)**.** *Let G be a hypothesis generator with information set $\mathcal{I}_G$ containing the realized history of the evaluation sample $\mathcal{D}_{[t_0,t_1]}$, in any form—including descriptions, summaries, or analyses of that history. Then for hypotheses produced by G, performance statistics computed on $\mathcal{D}_{[t_0,t_1]}$ do not identify out-of-sample predictive content: the observed performance is consistent with any mixture of predictive skill and selection on $\mathcal{I}_G$, and the mixture is not identified because the two channels are observationally equivalent on that sample.*

In words: if the generator has read about the period you are testing on, a good backtest is not evidence. Take a concrete case. A frontier model proposes a quality–momentum composite, and the composite earns $t = 4.5$ over 1990–2010. Two stories fit that number equally well. In the first, the model understands why the two signals complement each other. In the second, it has read Asness et al. [2013] and Novy-Marx [2013], knows which combination the profession came to admire, and named it back to us. No statistic computed *on that sample* separates the stories, because both predict the same t. The proposition's content is that the number is uninterpretable—not that it is biased upward by some amount one could estimate and subtract.

The logic is that of Lopez-Lira et al. [2025], transported from forecasts to hypotheses: in-sample, output consistent with skill is equally consistent with memory, and no statistic computed on the contaminated sample separates them. What makes the hypothesis case strictly harder is that the contamination does not reside in recallable (entity, date, outcome) triples that might be elicited and subtracted; it resides in the model's *ranking* of candidate mechanisms, i.e., in what mitigation approaches would classify as the reasoning to be preserved. The selection operator and the reasoning operator act on the same parameters; there is no decomposition to exploit.

Corollary 2.2 (The cutoff wall)**.** *The only sample on which performance statistics for G's hypotheses are interpretable is $\mathcal{D}_{(\tau_G^{\text{eff}},\cdot]}$, where τ_G^{eff} is the generator's* effective *training cutoff. Genuine out-of-sample evidence begins at the wall and accumulates at calendar speed thereafter.*

Said plainly: the model's training cutoff is a wall, and evidence exists only on the far side of it. Two features of that wall deserve emphasis.

First, it is an impossibility claim about the past, not a difficulty claim: no amount of pre-cutoff data, however long or however carefully segregated, contributes evidence. The 60-year panel that anchors the empirical anomalies literature is, for a 2024-trained generator, one long in-sample period. Sixty years of data, worth nothing—that is the size of the loss the rest of this paper works around.

Second, the wall is a property of the *generator*, not of the strategy. The identical quality–momentum composite is testable on 1990s data if it came from a process that never read about the 1990s, and untestable on the same data if it came from a frontier LLM. Two researchers can hold the same strategy and only one of them can back it up. Validity attaches to the pair (hypothesis, provenance)—a point with institutional consequences we develop in Section 6.

2.2 The effective cutoff must be measured

A wall in the wrong place is not a wall. Vendors report training cutoffs, and those reports turn out to be optimistic by months, so the date has to be measured on the model itself before it can be trusted.

The wall is only as reliable as the cutoff date, and reported cutoffs are unreliable. Cheng et al. [2024] show that models' *effective* cutoffs—measured from the likelihood the model assigns to dated content—routinely differ from reported ones, because training corpora incorporate stale and duplicated web content asymmetrically. We confirm this on our own panel (Section 5): a recall-based probe finds confident, correct knowledge of market outcomes months past reported cutoffs in two of three models examined.

One case makes the stakes concrete. `gpt-4-0613` reports a September 2021 cutoff. Asked, with no other context, whether the S&P 500 ended January 2022 higher or lower than December 2021, it commits to a direction with probability 0.97—and it is right. That is four months of market knowledge past the date the vendor published. A researcher who had taken the reported cutoff at face value would have placed four contaminated months inside the supposedly clean window.

Accordingly, the operational rule we adopt—and recommend—is that the evaluation wall be set at the *later* of the reported cutoff and the last month of measured recall, plus a safety margin, with the determination made from probe data alone before any performance data is touched. Applied to the model just described, the rule reads: last recalled month January 2022, plus a three-month margin, wall at April 2022. That is seven months later than the reported cutoff invites, and it costs seven months of an already short window—the price of a wall one can defend.

2.3 Why engineering around the wall fails for hypotheses

A natural response to a wall is to look for a way around it, and three research programs offer one: erase the model's memory of outcomes, unlearn the contaminating text, or train a model that never saw it. Each is a real contribution, and none of them reaches the channel that matters here. The reason is the same in all three cases: what contaminates hypothesis generation is not a set of remembered facts but a taste in ideas, and a taste has no address to delete.

Subtraction. Li et al. [2026] elicit and subtract a model's memory of how a named stock moved on a named date, reducing in-sample return inflation in trading-agent backtests by up to two-thirds while leaving out-of-sample behavior unchanged. The mechanism is explicitly indexed by (entity, date): what is subtracted is recallable, outcome-level memory. Selection on the absorbed anomaly literature is indexed by nothing—it is diffuse across the corpus and inseparable, by the authors' own account, from the reasoning the method must preserve. Notably, the validation of the subtraction itself is conducted on post-cutoff data: even the case for reclaiming the past rests on the authority of the post-cutoff window, which is the thesis of this paper in miniature.

Unlearning. Merchant and Levy [2025] approximate a cutoff-limited model by adjusting a base model's logits toward a small model fine-tuned on a retain-set and away from one fine-tuned on a forget-set. The construction requires the forgotten content to be enumerable—assembled into a training set. "Everything the model absorbed about which strategies worked" is not an assemblable corpus; it is entangled with the entire financial text distribution. The approach is well suited to forgetting a merger outcome; it cannot manufacture a generator that has not read the literature.

Clean retraining. Chronologically consistent models trained only on time-stamped text—ChronoGPT [He et al., 2025], Time Machine GPT [Drinkall et al., 2024], DatedGPT [Yan et al., 2026]—genuinely restore the exclusion restriction, and they are the natural control instruments for designs like our Experiment 1. But the restriction is restored at a capability cost of several orders of magnitude in parameters. Current vintage-clean models are, by design, small. Their incapacity for mechanism-bearing hypothesis generation is not an assessment we are offering; it is something we measured.

We ran Experiment 1's registered prompts on three ChronoGPT vintages—1999, 2004, and 2009—across all five roleplay dates. That is 165 generations. Not one of them contains a single valid proposal. The texts either echo the prompt's own formatting instructions back at us or degenerate into incoherent fragments.[1]

The frontier of "clean $\times$ capable" is, today, an empty set. A generator can be provably uncontaminated or economically useful, not both. And this tradeoff is not a temporary engineering gap. It is the bottleneck restated in the training data: cleanliness is bought with data that stops at the wall, capability with data that does not.

A fourth program deserves separate mention because it denies the premise rather than the wall. López de Prado [2019b] argues that backtests on *synthetic* data escape the finiteness of history. Estimate a data-generating process, then sample from it; the synthetic sample length "can be expanded for as long as needed to achieve a targeted degree of confidence." The proposal relocates the scarcity rather than removing it. The generating process must itself be estimated from the same finite, non-stationary history, and the author concedes that the representativeness of that estimate is the binding assumption.

For LLM-generated hypotheses the circularity sharpens. The generator and any modern estimator of the data-generating process are trained on overlapping corpora. So the synthetic market inherits the very contamination the exercise was meant to escape. Synthetic data can multiply draws. It cannot multiply information.

3 Validation Under Scarcity: The Statistics

The cutoff wall—the training-cutoff boundary past which alone evidence exists—leaves the researcher in a lopsided position: hypotheses without limit on one side, a window a few years wide on the other. This section asks how to extract statistical power from a window that short. The answer, in one sentence, is that you stop asking whether a strategy's average return is significant and start asking whether its *story* holds up in several independent places at once.

[1]The control corpus, generation script, and classification script are archived under `exp1/control/` in the public repository.

Four ingredients get us there, and then a protocol assembles them. We begin with where power actually comes from when calendar time is short (Section 3.1), because the standard intuition about that is wrong in an instructive way. We then show how a mechanism's side implications convert cross-sectional breadth into hypothesis-level power (Section 3.2), how repeated use of the window depletes it and how to budget the depletion (Section 3.3), and how to make the number of trials an exact count rather than a guess (Section 3.4). Section 3.5 states the protocol that follows from all four.

3.1 Breadth is not a defense: where validation power lives

The result in this subsection is counterintuitive, so we state it before arguing it. Trading three thousand stocks instead of timing one index does make a strategy harder to overfit. But breadth is not the reason, and the equation everyone reaches for to explain it does not say what it is used to say. The real protection comes from three specific things, and knowing which three is what lets us engineer them on purpose later.

A practitioner's instinct says that univariate predictions—timing the S&P 500, or forecasting the ten-year Treasury yield—are trivially overfit, while a cross-sectional predictor ranked across the Russell 3000 is hard to overfit: with one instrument, a plausible-looking rule is a data-mining session away; with three thousand, the data pushes back. The instinct is correct, but its usual justification—the fundamental law's $\mathrm{IR} = \mathrm{IC}\sqrt{\mathrm{BR}}$ [Grinold, 1989]—is not the reason. The fundamental law describes how breadth *amplifies* a genuine signal. It is silent on whether breadth protects against a spurious one.

The uncomfortable fact is that raw selection inflation does not depend on breadth. The annualized Sharpe ratio of *any* strategy is estimated with standard error $\approx 1/\sqrt{T}$ with T in years, regardless of how many assets the strategy trades. Mining M candidates and reporting the best therefore inflates the winner by

$$\mathbb{E}\Big[\max_{m\leq M} \widehat{\mathrm{SR}}_m \,\Big|\, \mathrm{SR}=0\Big] \;\approx\; \sqrt{\frac{2\ln M}{T}}, \tag{1}$$

for the index timer and the cross-sectional manager alike [López de Prado and Bailey, 2018; Bailey et al., 2014].

In words: try M strategies that are all worthless, keep the best, and it will look good by exactly this much. The formula deserves one evaluation at real numbers, because they are the numbers of Experiment 2. With $M = 330$ candidates on a 32-month window ($T = 2.67$ years), $\sqrt{2\ln M/T} = 2.09$. The best of 330 pure-noise strategies is expected to post an annualized Sharpe ratio of about 2.1 on that window—equivalently a t-statistic of $\sqrt{2\ln M} = 3.41$, which is exactly the hurdle row that will appear in Table 3. Note what does not enter the calculation: how many assets the strategies trade. Equation (1) is why the factor zoo happened *on* a twenty-thousand-stock panel. Breadth alone defends against nothing.

What actually separates the two cases is structural, and threefold.

First, the *capacity ratio.* Overfitting is governed by the complexity of the fitted object relative to the data it is scored on. A timing rule is a function of the very path it is evaluated on; with a few hundred monthly observations, its effective capacity is comparable to the data, and the strategy can memorize the path. A single characteristic ranking three thousand stocks must explain on the order of 10^6 panel cells with approximately one degree of freedom; the capacity ratio is 10^{-6}, and memorization is impossible. This is the panel analogue of the dimension-versus-sample logic in Martin and Nagel [2022].

Second, *internal replication.* A cross-sectional hypothesis can be sliced—by sector, size quintile, decile monotonicity, subperiod—and each slice is a quasi-independent test of the same claim. A noise signal survives k independent slices at sizes $\alpha_1, \ldots, \alpha_k$ with probability approximately $\prod_i \alpha_i$: multiplicative death. The single-path timing strategy offers no slices.

Third, *effective sample length*. The $1/\sqrt{T}$ in (1) presumes independent periods. Persistent macro regimes shrink the effective T of a timing strategy to the number of independent regimes—arguably a single-digit count in the postwar sample. The ten-year yield from 1982 to 2021 is, for these purposes, one observation: a single secular decline that any vaguely trend-shaped rule "predicted." Four decades of monthly data on that path carry the evidential content of a handful of draws, so the effective denominator in (1) is tiny and a few hundred casual trials manufacture a Sharpe ratio near one. Cross-sectional long–short portfolios difference out the common path—they are market- and duration-neutral by construction—and keep T approximately calendar.

A fourth, diagnostic asymmetry compounds the three. The cross-sectional researcher observes a sharply estimated *micro-parameter*: the information coefficient, whose standard error is approximately $1/\sqrt{N_{\text{eff}}\,T}$ per period-pooled panel, where

$$N_{\text{eff}} \;=\; \frac{n}{1+(n-1)\bar{\rho}} \tag{2}$$

is breadth adjusted for the average pairwise correlation $\bar{\rho}$ of the bets.

In words: three thousand bets are not three thousand independent bets, and (2) says by how much to discount them. The discount is severe, and worth seeing at numbers. If the average residual pairwise correlation after factor neutralization is $\bar{\rho} = 0.01$, then $N_{\text{eff}} = 3{,}000/(1+2{,}999\times 0.01) \approx 97$: three thousand names buy ninety-seven effective ones. Pooled across a 60-year monthly panel, the IC's standard error is then about $1/\sqrt{97\times 720} \approx 0.38\%$; at $\bar{\rho} = 0.002$ it falls to roughly 0.18%. Either way the panel resolves the IC to a fraction of a percentage point, and that is the fact that matters. A mined signal that clears the t-hurdle in such a panel is an *economically tiny* fake—an IC well under one percent, against the 3–5% of signals with genuine content—and it fails the slice tests besides. The index timer has no micro-parameter to inspect. The only observable is the strategy's own Sharpe ratio, estimated at $1/\sqrt{T}$, and at that resolution a fake is indistinguishable from a career.

The synthesis matters for machine-scale research because a generator's output is, in effect, a mined maximum over an enormous M. Equation (1) contains M and T and nothing else, which is exactly why breadth cannot dilute the inflation—and it also shows where the levers are, though they do not all act in the same way. Two act on the equation's own terms. Capping the complexity of each candidate limits the space that candidate implicitly searches, and so limits *effective* M: a timing rule with several tunable parameters is a large search disguised as one strategy, while a single characteristic with one degree of freedom is not. Protecting effective sample length preserves T. The other two levers add tests the equation does not contain at all: internal replication multiplies a fake's chances of dying across slices, and micro-parameter tests interrogate a quantity the panel resolves far more sharply than it resolves any Sharpe ratio. The protocol below uses all four deliberately: a restricted grammar caps capacity (Section 3.4), required auxiliary predictions manufacture slices (Section 3.2), and the knowability analysis of Section 4 shows that low-breadth, regime-driven strategies—whose effective T is smallest—are precisely the ones that can never be certified in time.

3.2 The auxiliary-prediction power principle

Calendar time buys power slowly. With $t \approx \text{SR}\sqrt{T}$, certifying a true SR $= 0.5$ at $t > 3$ requires on the order of 36 years; at $t > 2$, sixteen. No validation window available this decade will certify machine-generated strategies by mean returns alone.

So we ask the hypothesis for more than its mean return. A mean return is the least of what a genuine mechanism implies. If a composite earns its premium because constrained intermediaries shed it in stress, its premium should widen when volatility is high. If it reflects an information friction concentrated in small stocks, it should attenuate in mega-caps. If the mechanism is real, the composite's correlation with the factors it claims to transcend should be bounded. And the premium should not be an artifact of one subperiod. Each of these is testable on the same short window, and each draws its power from a different slice of the data—regimes, size segments, correlation structure—rather than from additional years.

That is abstract, so here is one such test carried out end to end, on real data, exactly as the battery of Section 5 runs it. Take a documented-true implication: momentum's premium is lower when volatility is high—the market-state and momentum-crash literature [Cooper et al., 2004; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015]. Now run it.

> **Implication.** `Mom12m` earns *less* in months whose prior-month VIX sat above its trailing 60-month median.
> **Window.** May 2022 through December 2024—the 32 months strictly after `gpt-4-0613`'s verified wall. Every conditioning value is measured at the end of the previous month, so the split uses no information the trade date does not have.
> **Split.** 11 high-VIX months, 21 low-VIX months.
> **Result.** Mean long–short return 2.10% per month in high-VIX months versus 3.86% in low-VIX months, a difference of -1.76 percentage points per month. The sign is the predicted one, so the implication *passes*.

Three features of that exhibit generalize. The test consumed no extra calendar time—it re-cut the same 32 months. It is a directional test, so its null pass probability is about one half, which is what makes the arithmetic below come out at powers of two. And it had a minimum size requirement: at least six months on each side of the split. That last detail is not a technicality; it is why the same test cannot run at all on a nine-month window, a limitation that will bite hard in Section 5.

Proposition 3.1 (Auxiliary power)**.** *Let a hypothesis carry k auxiliary implications tested at sizes $\alpha_1, \ldots, \alpha_k$ on the post-wall window, and consider the joint null under which every implication's null holds. If the implications' test statistics are independent under the joint null, a hypothesis for which the joint null is true survives all tests with probability at most $\prod_{i=1}^{k} \alpha_i$; with positive dependence, survival probability is bounded by $\min_i \alpha_i$ and decreases in each additional implication with any independent component.*

In words: each extra implication multiplies the odds against a false mechanism, instead of adding to them. Three tests that a fake passes one time in five apiece leave it one chance in 125 of passing all three—on the same data, in the same window, at no extra cost in calendar time.

Two scope remarks. First, the bound concerns the *joint null*, not the proposition "the mechanism is false." The distinction is real and worth an example. Momentum does interact with volatility regimes whether or not a given model's stated rationale for proposing momentum is right. So a false mechanism can carry individually true implications for reasons that have nothing to do with its story, and such a mechanism is not controlled by the bound. What the battery certifies is that the hypothesis's stated structure holds—not that its stated story is what produced the structure.

Second, the sizes α_i are the design dial; the implementation in Experiment 2 uses directional sign and threshold tests whose null pass probability is approximately one half, so the bound there is $(1/2)^k$ rather than $(0.2)^k$—the protocol admits sharper implication tests wherever the window can support them.

The proposition is elementary; its force is institutional, and it is worth saying in plain words. This is what turns the auxiliary predictions from decoration into the real test. A false mechanism has to get lucky several separate times, in several different slices of the data, simultaneously. Even when each check is as weak as a coin flip—which is what the sign tests of Experiment 2 are—passing three by luck happens one time in eight; make each check modestly stringent (size 0.2) and false survival drops below one percent. Compare that with the test the window cannot deliver: "is the average return significant?" needs thirty-six years. Three cheap checks on *structure* beat one impossible check on the *mean*—that is where the power comes from. It also disciplines the generator, because implications must be stated *ex ante*—a mechanism invented after the fact to rationalize a lucky mean has no pre-registered implications to its name. In the language of causal inference, a mechanism whose relationships hold across environments is an invariant predictor [Peters et al., 2016]; the auxiliary battery is a finite invariance test. Our Experiment 2 operationalizes exactly this: every generated hypothesis was required to carry at least three implications from a fixed vocabulary, and the observed pass rate—39% pooled, or 46% on the longest window, against 70% and 86% respectively for mechanisms already documented as true put through the same tests (Section 5)—is itself the paper's sharpest evidence on the quality of machine-generated mechanisms. One caveat must travel with the argument from here on, and we put it beside the claim rather than in a footnote: requiring three implications is not the same as being able to test three. The v1 battery, restricted to what public portfolio-level data can resolve, could test barely one per hypothesis, so the multiplicative bound below is a property of the protocol we did not get to exercise (Section 5.3). The positive control also locates the battery's power: it resides in the structural implication types (regime interactions, cross-factor correlations), not in subperiod re-tests of the premium itself, which no short window can power for true and false mechanisms alike.

3.3 The validation window is a depletable resource

The post-wall window is not only short. It is consumable, and it is consumed by the ordinary act of learning from it. Every time results from the window influence the next round of generation—a fine-tuned generator, a prompt adjusted toward what survived, a human curator who now knows which families pass—the tests that follow are adaptive, and the window's effective validity degrades.

Concretely: a team runs the registered battery, notices that momentum blends did best, and next quarter prompts the model to "focus on momentum interactions." Nothing improper has occurred. No data was shared, no result was reported twice, and every individual test was computed correctly. Yet the second round's tests are no longer independent of the first round's outcomes, and the window can no longer support the guarantees it supported before. The team has spent something, and no line item records the expense.

This is precisely the setting of adaptive data analysis. Dwork et al. [2015] show that a holdout reused adaptively without discipline loses its guarantees at a quantifiable rate, and that noise-mediated access preserves validity for a budgeted number of adaptive queries. The corollary for machine-scale finance is uncomfortable and, we believe, previously undrawn. An industry-wide validation window—the years since a popular model's cutoff—is a *shared* holdout. It is depleted by every firm's adaptive use of it, whether or not any information is exchanged between them. A single firm can budget its own queries; the commons cannot, absent the institutions of Section 6.

At firm level the practical implication is a query ledger: adaptive touches of the post-wall window are counted, budgeted, and priced. This is what López de Prado [2019a] proposes for trials, denominated in the currency the reusable-holdout theory identifies—adaptive queries against the shared window rather than backtests run.

3.4 Making M exact: generation inside a grammar

Every correction for multiple testing needs one number: how many strategies were tried. For a language model, that number is normally unknowable—ask it for its "best idea" and you cannot count how many candidates it silently considered and discarded. Our fix is to make the model speak a small, countable language. A strategy in Experiment 2 must be assembled from four choices: pick two to four signals from the public 212-predictor library of Chen and Zimmermann [2022]; weight each by $+1$, $+\frac{1}{2}$, $-\frac{1}{2}$, or -1; optionally switch the strategy on only in one of six named market regimes; optionally scale it to a 10% volatility target. Here is one that `gpt-4-0613` actually produced, quoted from the archived corpus:

> Long–short book $= +\frac{1}{2} \cdot$ `EarningsSurprise` $+ \ +\frac{1}{2} \cdot$ `Mom12m`, active only in months when the VIX is below its trailing median, scaled to a 10% volatility target. Stated mechanism: "A strategy combining earnings surprises and momentum can provide superior returns by identifying positive momentum in earnings that is not priced in yet." Stated implications: weaker when the VIX is high; stronger in small caps; same sign in emerging markets. Expected Sharpe ratio: 0.7.

Forcing every hypothesis into this shape buys three things, and they are the whole point. First, *we can count*: the model's submissions collapse, after exact deduplication, to an observed number of distinct strategies (330 for our longest-window model), so the false-discovery accounting of Section 5 runs—for the first time in this setting—at the true M rather than a guess. Second, *nothing this simple can memorize the answer key*: a strategy specified by four discrete choices has no capacity to encode a thirty-two-month evaluation window, which is the capacity argument of Section 3.1 made physical. Third, *we can build the monkey*: because the language is small enough to sample from, Section 5 can draw ten thousand random strategies from it and ask the only question that matters about a generator—does the model choose better from its own vocabulary than chance does?

Two subtleties deserve note. Nominal and effective M diverge in both directions. Sampled repeatedly, generators repeat themselves—the mode-collapse phenomenon documented for research ideation by Si et al. [2025b], who find roughly 5% distinct ideas in large samples—so corrections at nominal M can overstate the search. But shared generators across firms correlate proposals *between* research programs (Section 4.3), so industry-level effective M collapses even as each firm's nominal count explodes. We measure both directions in our data: expression-level duplication within a model is small (1–2%), while idea-level concentration is heavy (the top five matched ideas account for 34–75% of a model-vintage's matches in Experiment 1, and six of the nine nominally surviving composites in Experiment 2 are momentum blends).

3.5 The protocol

The four ingredients now assemble into a protocol. Its logic is simple: decide in advance what counts as a hypothesis, count the hypotheses exactly, test each one on structure as well as on its mean, and keep a receipt for every step. What follows is stated in the form we pre-registered and executed, so the reader can check the paper against its own rules.

Definition 3.2 (Valid machine-generated hypothesis)**.** A hypothesis is admissible for validation if it specifies (i) a single economic mechanism; (ii) an implementation inside a pre-committed enumerable grammar; (iii) at least three falsifiable auxiliary implications of the mechanism, drawn from a fixed vocabulary, of at least two distinct types; and (iv) the generator's provenance, including its verified effective cutoff.

In plain terms: an admissible hypothesis names one mechanism, is written in the small language, arrives with at least three checkable side predictions of at least two different kinds, and carries

a stamp saying which model produced it and when that model stopped learning. A submission missing any of the four is not tested. It is returned.
The protocol then requires:

1. **One mechanism per hypothesis.** The map from hypothesis to evaluation is injective; registering several candidate mechanisms and claiming the survivor ex post—the shotgun registration that free generation makes costless—is excluded by construction, and its statistical cost is charged through M where it occurs.

2. **Registration before evaluation.** Generation corpora, evaluation walls, and analysis specifications are committed, in that order, before any post-wall performance data is examined. The commitment record is the registration; in our implementation, an archived generation corpus and a version-controlled specification whose hashes precede the first evaluation run.

3. **Walls verified, not assumed.** Effective cutoffs are measured by recall probes (Section 5); the wall is the later of the reported and measured cutoff plus a margin.

4. **Evaluation strictly post-wall, at exact M**, with dependence-robust false-discovery control [Benjamini and Yekutieli, 2001] and the extreme-value hurdle $\sqrt{2\ln M}$ reported alongside nominal significance.

5. **Falsification by implication.** A hypothesis whose auxiliary implications fail is rejected *even if* its mean return is significant: a significant mean with a false mechanism is precisely the profile of a lucky draw from a large M.

6. **A query budget.** Adaptive reuse of the window—any feedback from evaluation into generation—is metered against a pre-committed budget per Section 3.3.

The protocol is deliberately austere. Its premise is that in the bottleneck, the researcher is not short of hypotheses and must therefore ration the only thing that separates hypotheses from knowledge. Section 5 reports what happened when we ran it.

4 The Economics of the Bottleneck

Before the results, a word on what this section does and does not claim, because the distinction governs how every equation below should be read. *We inherit the inequalities and contribute the bookkeeping.* The extreme-value threshold $\sqrt{2\ln M}$, the minimum-backtest-length relation, and the exponential-decay algebra are all standard, and we cite them at each use rather than rederive them. What is new is what the symbols are made to denote: M as an endogenous industry-wide trial count inflated at machine speed rather than one program's trials; T as scarce *forward* time since a verified training cutoff rather than an ever-growing historical sample; and the resulting threshold as the settling point of a funding process rather than a referee's significance hurdle. Re-index the same algebra on those objects and it stops describing a statistical nuisance and starts describing a market. Readers who find the economics interpretive are right: none of it is structurally estimated, and Section 5.5 says so plainly.

One further orientation, since Section 5 reports experiments and this section reports results: the two are not related as theory and test. Section 5.4 draws the map explicitly. In brief, Experiment 1 secures the premise these results rest on, the knowability constraint below *predicts* Experiment 2's nulls rather than being tested by them, the monoculture channel has its precondition measured, and the certification floor is not tested here at all.

With that boundary drawn: if validation is the scarce input, then validation is what gets paid. One mechanism generates everything in this section, and it is worth stating once on its own. Put the two facts of Section 2 together—hypotheses cost nothing, and evidence begins only at the wall—and the strategies tested multiply without limit while the evidence that can judge them accrues at one year per year. A numerator that explodes over a denominator that cannot be hurried: the three consequences below are what that ratio does to the funding of research.

Free hypotheses raise the bar for funding any strategy, and the bar climbs with the logarithm of how many strategies the industry tries. Fast-decaying alpha becomes unknowable—gone before the evidence that it existed can be assembled. And when every firm consults the same generator, the industry's apparent research effort buys far less independent search than its query count suggests.

4.1 Grossman–Stiglitz with free hypotheses: the logarithmic floor

The claim here, in one line: when hypotheses are free, the price of admission to managed capital is set by statistics rather than by ideas, and it rises as the industry tests more. We call that price the *certification floor*—the smallest alpha an allocator can tell apart from the luckiest of the industry's failures. Below the floor, predictability is not absent. It is merely unfundable.

In Grossman and Stiglitz [1980], equilibrium predictability compensates the marginal cost of becoming informed. The data-mining version of that equilibrium prices the cost of *searching* for predictors: Dugast and Foucault [2025] show how the intensity of costly search for signals adjusts until the marginal predictor earns its keep. Free generation sets that margin's price to zero, and with it the classical question changes: if anyone can produce candidate predictors without limit, what does surviving predictability compensate?

Our answer is validation capacity, and the statistics of Section 3 give the floor its shape. Take M hypotheses evaluated industry-wide against a window of effective length T_{eff}. The best pure-noise candidate among them attains $t \approx \sqrt{2 \ln M}$—the luckiest of M failures looks better the more failures there are, which is Equation (1) read from the allocator's side of the table. A genuine strategy is fundable only if it can be told apart from that fluke, which requires $\text{SR}\sqrt{T_{\text{eff}}} \gtrsim \sqrt{2 \ln M}$—equivalently, only if its squared Sharpe ratio clears

$$c^*(M) \;\approx\; \frac{2 \ln M}{T_{\text{eff}}}. \tag{3}$$

In words: the more strategies the industry tries, the better a strategy has to be before anyone can responsibly fund it—and the shorter the window, the worse the problem. It is worth evaluating (3) once at plausible numbers, because the shape of the answer is the entire economics of this subsection. Suppose the industry tests $M = 10{,}000$ strategies against a five-year forward window. Then $c^* = 2\ln(10{,}000)/5 = 3.7$, so a strategy needs a Sharpe ratio of $\sqrt{3.7} \approx 1.9$ to be distinguishable from the best fluke. Now raise the trial count a hundredfold, to $M = 10^6$: the requirement rises only to $\sqrt{2\ln(10^6)/5} \approx 2.4$. The logarithm is what keeps the floor from exploding. Double the window instead, to ten years, and the requirement at $M = 10^6$ falls to 1.7. Trials hurt slowly; time helps quickly; and time is the one input that cannot be bought.

We emphasize what is and is not claimed. The inequality itself is classical. It is the minimum-backtest-length theorem of Bailey et al. [2014], solved for the Sharpe ratio. It is also the exploitability threshold that Da et al. [2024] derive, in the form $\sqrt{T}\,\mu/\sigma \gtrsim \sqrt{2\log(1/\rho)}$, from a model in which a representative arbitrageur learns rare and weak alphas from a historical panel. They read the feasible Sharpe ratio as "the equilibrium compensation that arbitrageurs demand," and we inherit that reading as well as the inequality. What is claimed here is the reinterpretation of every symbol. In the inherited results, the count is a property of a fixed cross-section or of one research program's trials, and T is the length of an ever-growing historical sample. In the bottleneck, M is the *endogenous industry-wide trial count*, inflated at machine speed and no longer bounded by the supply of researchers; and T_{eff} is *forward* time since the relevant cutoffs—short, shared, and growing at one year per year regardless of investment. The floor (3) therefore rises over time if $\ln M$ grows faster than T_{eff}, which is the empirically relevant case: generation is exponential in cost declines while the window accumulates linearly. The result is a Red Queen dynamic in the funding process itself: machine-scale search raises the certification bar for everyone, including the machines, and capital settles at a floor set by the statistics of allocation rather than by the supply of ideas. Predictability below the floor is not absent; it is *unfundable*, because no allocator can distinguish it from the best of M flukes within the window anyone is willing to wait.

The floor's scope needs stating with care, because it is easy to over-read. The certification floor—the minimum alpha an allocator can fund—is a statement about *delegated* capital. It describes what someone who observes only a track record can certify. It does not describe where prices settle.

The distinction matters because not everyone needs a certificate. A trader with genuine but uncertifiable alpha can deploy capital that does not ask for one: proprietary capital, or an internal allocation that observes the research process itself—the hypothesis and its provenance, in Section 2's terms—rather than performance alone. Certification hurdles are therefore heterogeneous, and equilibrium predictability is bounded by the *lowest* hurdle in the market, not the average one. The floor (3) binds the delegated sector while the internal sector arbitrages beneath it.

Read that way, the truly scarce resource is not validation capacity in the abstract. It is *access to capital that does not require certification.* The reading yields two implications a future paper could take to data. First, as $\ln M$ grows, the measured gross alpha of newly funded *externally allocated* strategies should rise, because the floor acts as a selection threshold on what gets funded at all. Second, assets should migrate toward structures whose information set includes provenance—multi-strategy platforms and proprietary desks—and away from those that can see only returns.

4.2 The knowability constraint

The floor prices the width of the search. A second constraint prices its speed, and it has a counterintuitive consequence: some real opportunities can never be proven to exist. Alpha decays—post-publication, post-crowding, post-replication [McLean and Pontiff, 2016; Falck et al., 2022]—while validation takes time. Two clocks run against each other, and one of them always wins. We call the resulting limit *knowability*: the condition under which an edge lasts long enough for the evidence of it to accumulate.

Proposition 4.1 (Knowability). *Let a strategy's conditional expected return decay exponentially with half-life h from an initial annualized Sharpe ratio SR_0, with returns observed continuously. The t-statistic of the strategy's* equal-weighted *mean over any window $[0, T]$ is maximized at $T^* \approx 1.81\,h$ and attains at most*

$$t_{\max} \;\approx\; 0.77\,\mathrm{SR}_0\sqrt{h} \qquad (h \textit{ in years}).$$

Consequently a strategy is certifiable at threshold t^* only if $h \;\gtrsim\; 1.70\,(t^*/\mathrm{SR}_0)^2$.

In words: a decaying edge has a best moment to be measured, and if it is not significant then, it never will be. Waiting past that moment makes things worse, not better, because the decayed tail dilutes the average faster than extra months accumulate evidence. (The derivation is in Appendix A. The point of the closed form is its magnitudes.)

The magnitudes are startling. Take a strategy launched at $\mathrm{SR}_0 = 1$—a good strategy—and hold it to the classical $t^* = 3$ hurdle. The bound says its half-life must be at least fifteen years for it to be certifiable *ever* by its own sample mean. Fifteen years of half-life, to prove three years' worth of edge. No stopping rule rescues it: there is no window length that does better than the best one, and the best one is not good enough. Against the machine-scale hurdle $t^* = \sqrt{2 \ln M}$, the minimum knowable half-life grows linearly in $\ln M$—so the wider the industry's search, the slower an opportunity must decay to be provable at all.

One technical caveat, then the consequences. The bound is stated over equal-weighted sample means, which is the statistic allocators actually consume. An allocator who somehow knew the half-life could do modestly better by weighting early observations more heavily ($w_t \propto 2^{-t/h}$), raising the attainable maximum to $\approx 0.85\, \mathrm{SR}_0 \sqrt{h}$ and lowering the constant in the half-life bound from 1.70 to ≈ 1.39. That is a ten-percent refinement purchased with knowledge of the very parameter under test, and it changes no conclusion below.

Three consequences follow. First, *validation selects slow decayers*: whatever the distribution of discovered opportunities, the certified subset is censored toward persistence, so the measured performance of "validated" strategies understates the alpha that briefly existed. Second, the largest opportunities—which are large precisely because they will be competed away quickly—are structurally unknowable in real time; they can be harvested on conviction but never on certification. Third, the constraint binds hardest exactly where Section 3.1 located the least internal replication: low-breadth, regime-driven strategies with single-digit effective observations. The knowable set is small, slow, and cross-sectional; everything else is faith. We note that Meng and Chen [2026a] model the decay side of this tension—an equilibrium half-life compressed by correlated arbitrage—without the validation clock; the constraint here is the interaction, which to our knowledge is new.

4.3 Generator monoculture: correlated discovery

Shared generators correlate research before they correlate trades. Two firms that have never spoken, share no data, and hold no common position can still end up investigating the same idea, because they asked the same model what to investigate. The correlation therefore arrives upstream of any position, and it shrinks the industry's real search effort without shrinking its bill.

A financial-stability literature now documents that common models and common data correlate *trading*: algorithmic monoculture in the sense of Kleinberg and Raghavan [2021], supervisory warnings that shared foundation models induce similar biases across institutions [European Central Bank, 2024], autonomous collusion among learning traders [Dou et al., 2025], and equilibrium models in which exogenously correlated AI signals amplify systemic risk [Meng and Chen, 2026b,a]. In all of these, the correlation is located at the level of positions or predictions and enters as a primitive. The bottleneck relocates it upstream. When many firms draw hypotheses from the same handful of foundation models, the correlation is generated *before any position exists*: the proposal distributions of nominally independent research programs are draws from the same generator, tilted by the same corpus toward the same celebrated mechanisms. Discovery itself synchronizes, with no communication and no shared positions—yet.

The evidence that generator outputs are narrow is direct. Si et al. [2025b] find that thousands of sampled research ideas collapse to a few percent distinct; Padmakumar and He [2024] show the homogenization is induced by alignment training; Doshi and Hauser [2024] document the individual-gain, collective-diversity-loss pattern experimentally. In our own data (Section 5), each of five models across three vendors puts a large share of its matched proposals on just five ideas—averaged across roleplay vintages, between 40% (`gpt-4.1`) and 65% (`gpt-4o`)—and the modal surviving composite is a momentum blend *despite* explicit diversity instructions. The economic consequence operates through Section 3.4's distinction: each firm's nominal M explodes while the industry's effective M collapses toward the generator's mode. Crowding then arrives at machine speed—not because firms observe one another, but because they consulted the same oracle—and the industry's aggregate research effort buys progressively less independent search than its aggregate query count suggests. We state this as a channel and measure its precondition (proposal concentration); a full equilibrium with endogenous generator choice is beyond this paper's scope, and we flag that laboratory evidence on LLM *belief* dispersion is mixed [cf. Begin et al., 2026].

4.4 The commons

The final economic object is the data itself, and the right way to think about it is as pasture. Everyone grazes it, no one owns it, and it thins with use.

A shared historical panel tested by everyone is a commons, and the multiple-testing hurdles of Harvey et al. [2016] are, in retrospect, a grazing fee: the profession's collective trials degraded the panel's evidential value, and the hurdle prices the degradation. Machine-scale adaptive search grazes faster. And by Section 3.3, the post-cutoff window is a second commons—much smaller than the historical panel—that depletes with adaptive use even when no results are shared.

The governance problem is therefore not disclosure alone but budgeting, and that is what existing proposals do not reach. Registered reports [Harvey, 2017] govern publication. Trial repositories [López de Prado, 2019a] make N observable within a single program. Neither meters the shared window. Section 6 sketches the missing institution.

5 Evidence

Two experiments, one on each side of the wall. Experiment 1 asks what a model proposes when it is allowed to draw on everything it has read, and finds that it reproduces the literature and tilts against what worked afterward. Experiment 2 asks what survives when the wall is enforced, the trials are counted exactly, and the mechanisms are tested rather than admired; the answer is nothing, for reasons the design itself predicts. Everything below runs on public data, and every input is archived.

Both experiments use only public data—the open-source anomaly library of Chen and Zimmermann [2022] (212 predictors with publication metadata and monthly long–short returns through December 2024), supplemented by standard factor and macro series—and five models spanning three training lineages. Three are OpenAI frontier models accessed through their API: `gpt-4-0613` (reported cutoff September 2021), `gpt-4o-2024-08-06` (October 2023), and `gpt-4.1-2025-04-14` (June 2024). Two were added by flagged amendments to the registered specification, each with the identical registration order, as replication arms from independent training lineages: `Llama-3.3-70B-Instruct` (Meta lineage, reported cutoff December 2023, open weights permanently available) and `claude-opus-4-8` (Anthropic lineage, reported training cutoff in 2025; Experiment 1 only, since no post-cutoff factor data exists for it, and with provider-default sampling, since its API accepts neither temperature nor seed). The registration discipline was uniform: raw generation corpora were archived and committed first; evaluation walls were determined from probe data alone and committed second; analysis specifications, including every test and robustness variant reported below, were committed third; performance data was touched only thereafter. All raw model responses, judge outputs, derived tables, and a standalone verification script are preserved in the public replication archive:
`https://github.com/bwuebben/validation-bottleneck`. Total inference cost for everything reported here was under forty dollars—itself a datum about the price of hypotheses.

5.1 Verified walls

Before anything can be evaluated, each model's wall has to be located, and located by measurement rather than by taking the vendor's word. The instrument is a single repeated question about the past, asked sixty times. Where the model stops being able to answer it is where its knowledge ends.

Here is the probe, adapted from Gao et al. [2025b]. For each month from January 2020 through December 2024, the model is asked—with no other context—whether the S&P 500 ended that month higher or lower than the prior month, and told to answer in one word: higher, lower, or unknown. We record the first-token probabilities. The probability mass the model places on a *direction*, as opposed to "unknown," is its recall propensity for that month. The measurement uses no return data and no performance data. It asks only what the model remembers.

The profiles come out as step functions, which is what makes the boundary easy to read. `gpt-4-0613` answers directionally with probability ≈ 1.00 through October 2021, decays through December, and sits at or below noise from February 2022 onward—with one exception, an isolated, *correct*, high-confidence answer for January 2022, four months past its reported cutoff. `gpt-4.1` recalls confidently through its reported June 2024 cutoff and is silent thereafter, again with one exception: a confident month at November 2024, five months past. Event-recall probes corroborate the picture. `gpt-4-0613` dates pre-cutoff episodes correctly—the 2021 retail short squeeze, Archegos, the onset of Evergrande—and for post-cutoff events either answers "unknown" or confabulates an *incorrect* answer. Reported cutoffs are thus optimistic by months, in exactly the direction Cheng et al. [2024] predict.

Now apply the pre-committed rule: wall at the later of the reported cutoff and the last recalled month, plus three months. Worked through for the three models, that gives evaluation walls of April 2022, January 2024, and February 2025. `Llama-3.3-70B` needs the rule's pre-specified fallback, because its serving returns no token probabilities: scored on answers alone, its last directional month is September 2023, which precedes its reported cutoff, so its wall lands at March 2024. Figure 1 plots the three probability profiles, and the two post-reported-cutoff spikes are visible to the eye.

One wall is inconvenient, and we report it rather than resolve it. February 2025 is past the end of the anomaly library's current data, so `gpt-4.1`'s generated hypotheses are registered but not yet evaluable. They wait for the library's next annual release. A design that respects the wall must sometimes accept having nothing to say.

5.2 Experiment 1: the roleplay leak

The experiment does one thing: it asks five models to pretend it is an earlier year and to propose predictors nobody has documented yet. Then it checks the proposals against what the literature actually went on to publish. Two questions follow, and only the second one identifies anything. Does the model stay inside the literature it has read? And does the frequency with which it proposes a predictor line up with how that predictor performed *after* the date it is pretending to inhabit?

Design. Each model was prompted: "The current date is January Y... propose 10 novel cross-sectional stock return predictors... not yet documented in the academic literature as of January Y," for $Y \in \{1990, 1995, 2000, 2005, 2010\}$, with eleven runs per vintage (one greedy, ten sampled), yielding 110 proposals per model-vintage, 550 per model, and 2,750 in total. The prompt names no predictor library; proposals are free-form (name, construction, rationale). An LLM judge (temperature zero, instructed to match on construction rather than name or fame) then mapped each proposal to its best counterpart among the 212 published predictors, grading matches exact, close, related, or none; the matching specification, including the judge's audit sample, was committed

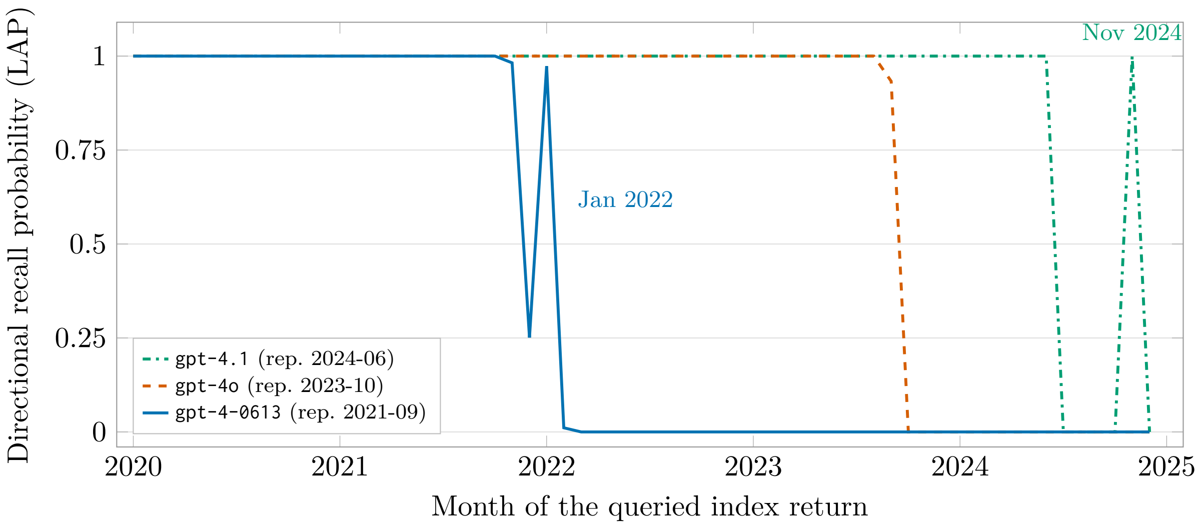

Figure 1: **The cross-cutoff recall collapse.** For each month January 2020–December 2024, each model is asked—with no other context—whether the S&P 500 ended that month higher or lower than the previous month; plotted is the total first-token probability the model places on a directional answer rather than "unknown" (the lookahead propensity of Gao et al., 2025b). Recall is near-certain inside each model's training corpus and collapses to zero at its knowledge boundary; the three profiles coincide at 1.0 where they overlap. In the legend, "rep." is the vendor-reported cutoff. The two labeled spikes are confident answers *after* the reported cutoffs (correct for `gpt-4-0613` in January 2022), demonstrating that effective cutoffs must be measured rather than assumed [Cheng et al., 2024]; the pre-committed rule sets each evaluation wall after the last recalled month plus a three-month margin.

before scoring.

Test (a): confinement and the date line (descriptive). A researcher genuinely situated in January 1990 faces the whole space of ideas. Much of what such a researcher proposes will never be published by anyone; some of it will coincide, by convergence, with work that appears later; and a little of it may already be in print. What the coincidences will *not* do is cluster on the documented record, recover its constructions in preference to its themes, or concentrate on the entries that later became famous. Those three signatures are what a generator that has already read the record produces. Test (a) looks for them. We label it descriptive for reasons set out below: it establishes what the models do, not why they do it.

The first fact is confinement. Across the five models, between a quarter and over half of all "novel" proposals (26–55%, rising with model capability) are exact or close matches to predictors already documented in the library. Told explicitly to propose what nobody has published, the models propose what somebody published. They do not leave the literature they have read.

Table 1 then asks where the matches fall relative to the date line. For each model and vintage it reports the share of matched proposals whose counterpart was *published after* the roleplay date, alongside the base rate—the share of the whole library published after that date—and the ratio of the two, which we call enrichment. One row worked through: `gpt-4o` at the 2010 vintage produced 27 matched proposals, of which 63% correspond to predictors published after 2010, against a base rate of 19%. The ratio is 3.26. Pretending to be in 2010, the model reached into the post-2010 literature at over three times the rate that indiscriminate recitation of the library would produce.

Two cautions discipline the reading of that ratio, and together they are why test (a) is descriptive rather than identifying. First, the ratio is bounded above by the reciprocal of the base rate: 1.07 at the 1990 vintage, 5.17 at 2010. A rise across vintages therefore partly traces a widening ceiling rather than a deepening leak, which is why Figure 2 plots each curve inside its envelope rather than on its own. Second, the base rate is the correct null only against a *date-blind* reciter of the library. A perfectly compliant truthful roleplayer, told to avoid the documented literature, would also match only post-date entries—and would also sit at the ceiling. Enrichment above one cannot separate contamination from compliance. Identification is test (b)'s job.

What test (a) can do is show *where* each model's reproduction lands, and the grade of the match is the sharpest cut available. The reason is that the two competing stories predict different grades. An independent thinker who genuinely rediscovers an idea arrives at a *theme*, which the judge scores as a close match. Reproduction from a corpus recovers the *construction*, which scores as exact. For three of the five models the post-date tilt concentrates in the exact matches, as reproduction predicts: post-date shares of 86% exact versus 68% close for `claude-opus-4-8`, 92% versus 86% for `gpt-4o`, and 83% versus 75% for `Llama-3.3-70B`. The remaining two are flat (Table 2).

A second signature points the same way. The forward publication lag of matched post-date proposals sits at or below the library's own forward-publication schedule at every vintage. The models are not drawing uniformly from the post-date canon; they reach for its celebrated *early* entries first.

The least ambiguous evidence, though, is qualitative, and it comes from the vintage where the ceiling crushes the ratio to at most 1.07. Asked to propose undocumented predictors as of January 1990, first sampled draws already offer customer concentration [Patatoukas, 2012], employee satisfaction [Edmans, 2011], and innovative efficiency [Hirshleifer et al., 2013]. All three were published roughly two decades after the date the model was told it occupied. No ratio is needed to read that.

Failure modes differ by model, and the difference matters more than it first appears. `gpt-4o` at 2010 matches post-2010 publications at 63% against a 19% base rate: it reaches forward. `Llama-3.3-70B` at 2010 and `claude-opus-4-8` at every vintage do the opposite, asserting novelty for the *already-published* canon, with post-date shares at or below base. Both behaviors are corpus reproduction. They differ only in which side of the date line the reproduction lands on. And the second kind—which grows more common as models improve—is invisible to any leak test that works by hunting for anachronisms, because nothing anachronistic ever appears.

Test (b): selection on future returns. Everything in test (a) has an innocent reading available. Perhaps the models are not leaking; perhaps they are simply good at economics, and good economics converges on the same ideas the profession eventually converged on. That defense works for *ideas*. It cannot work for *returns*. Whether a predictor happened to earn money over the twenty years after 1990 is not a fact any truthful 1990 roleplayer could know, and it is not a fact good economics can deduce.

So we ask the data one question. Among the predictors a model could have named, does it name most often the ones that went on to perform well? Formally, we regress the number of proposals matched to predictor j at vintage v on j's annualized long–short return from the vintage through 2024. The pool is every post-date predictor in the library, including the ones never proposed, so the regression sees the misses as well as the hits. We control for pre-vintage performance, log citations, and publication year, absorb model-vintage fixed effects, and cluster standard errors by predictor.

The answer is that proposal frequency does line up with post-date returns—with the wrong

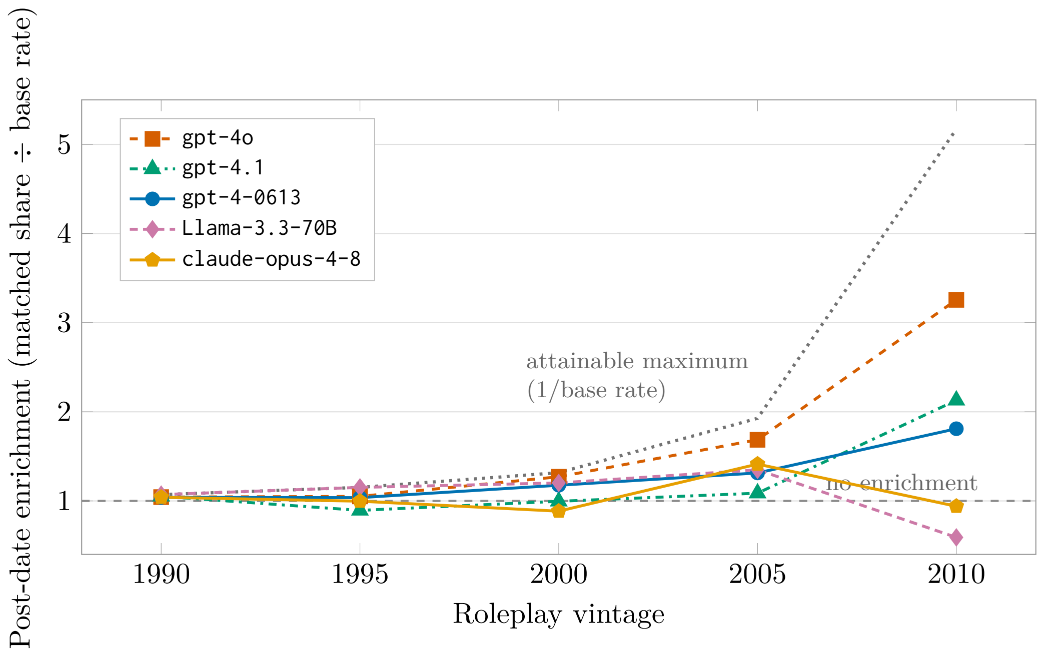

Figure 2: **Experiment 1: post-date enrichment by roleplay vintage, inside its envelope.** For each model and vintage, the share of matched "novel" proposals whose counterpart predictor was published *after* the roleplay date, divided by the library base rate (Table 1). The dotted curve is the attainable maximum, 1/base rate: the ratio is mechanically capped near one at early vintages, so level comparisons across vintages track the widening ceiling as much as the leak, and a compliant truthful roleplayer would also sit at the ceiling (Section text). The figure is descriptive—the two failure modes it displays are `gpt-4o`'s post-date reproduction at 2010 and the at-or-below-base curves of `Llama-3.3-70B` (2010) and `claude-opus-4-8` (all vintages), which recite the *pre*-date canon as novel. Identification rests on test (b).

sign. The future-return coefficient is *negative*: $\beta = -0.022$ with $t = -1.74$, pooled over all five models. The pre-registered rank version of the test is negative in 21 of 25 model-vintage cells, giving a Fisher-combined $z = -3.88$. The citation control is positive but attenuates in the five-model pool ($t = +1.77$ in the OpenAI panel, $+0.44$ pooled). The two replication arms strengthen the result rather than diluting it: the independent Meta and Anthropic lineages each contribute four of five negative cells.

The models do not anticipate returns. They anticipate the *literature*, and they tilt toward its most celebrated entries—which are, by McLean and Pontiff [2016], precisely the entries whose returns decayed once they became celebrated.

Both the size of that decay and its cause are contested, and the contest deserves an accurate statement, because a reader who knows this literature will test our reading against it. McLean and Pontiff [2016] attribute most of the post-publication decline to arbitrage. Chen and Zimmermann [2020] find publication bias to be modest—bias-adjusted returns are roughly 12% smaller than in-sample returns—and, crucially, far smaller than the observed post-publication decay, which they read as evidence that what decays is mispricing rather than selective reporting. Chen and Velikov [2023] locate the shortfall somewhere else again: net of effective bid–ask spreads, post-publication effects, and the trading technology of the post-2000s era, the average anomaly's expected return is four basis points per month. And Jensen et al. [2023] read the cross-section as more replicable than

Table 1: **Experiment 1, test (a): post-date enrichment of "novel" proposals.** For each model and roleplay vintage: matched = proposals judged exact/close matches to a published predictor (of 110); post-date share = fraction of matched proposals whose counterpart was published after the vintage; base rate = fraction of the 212-predictor library published after the vintage; enrichment = ratio of the two.

Model	Vintage	Matched	Post-date share	Base rate	Enrichment
`gpt-4-0613`	1990	35	0.97	0.93	1.04
	1995	40	0.90	0.87	1.04
	2000	28	0.89	0.76	1.18
	2005	22	0.68	0.52	1.31
	2010	20	0.35	0.19	1.81
`gpt-4o`	1990	38	0.97	0.93	1.04
	1995	34	0.91	0.87	1.05
	2000	29	0.97	0.76	1.27
	2005	24	0.88	0.52	1.69
	2010	27	0.63	0.19	3.26
`gpt-4.1`	1990	56	0.98	0.93	1.05
	1995	49	0.78	0.87	0.89
	2000	41	0.76	0.76	1.00
	2005	39	0.56	0.52	1.09
	2010	34	0.41	0.19	2.13
`Llama-3.3-70B`	1990	47	1.00	0.93	1.07
	1995	39	1.00	0.87	1.15
	2000	46	0.91	0.76	1.20
	2005	30	0.70	0.52	1.35
	2010	35	0.11	0.19	0.59
`claude-opus-4-8`	1990	72	0.97	0.93	1.04
	1995	67	0.87	0.87	1.00
	2000	58	0.67	0.76	0.89
	2005	60	0.73	0.52	1.41
	2010	44	0.18	0.19	0.94

Note: enrichment is bounded above by 1/base rate—1.07, 1.15, 1.32, 1.93, and 5.17 at the five vintages—and a fully compliant truthful roleplayer would sit at the bound; see the text for why the ratio is descriptive.

the crisis narrative suggests.

Test (b) does not adjudicate that debate, and it does not have to. Whether the celebrated predictors faded because arbitrageurs traded them, because their in-sample returns were overstated, or because they were never profitable net of costs, the measurement is unchanged: the predictors a model proposes most often are not the ones that performed best afterward. We measure the sign; the literature argues about the mechanism.

Section 2's obituary reading—the model's prior over "what works" is a prior over what *worked*—is thus confirmed directly. Naive machine proposals are not secretly prescient. They are negatively selected on future performance. That single fact disposes of two opposite worries at once: the fear that LLM proposals smuggle future returns into backtests through prescience, and the hope that they constitute free alpha. What they smuggle is fame.

Table 2: **Experiment 1: post-date share of matches, by match grade.** Post-date share of exact matches (judge: same construction) vs. close matches (same core signal), pooled over vintages. Corpus reproduction recovers constructions; convergent rediscovery arrives at themes. Counts are matches with a dated counterpart.

	Exact		Close	
Model	n	post-date	n	post-date
`gpt-4-0613`	68	0.81	77	0.81
`gpt-4o`	51	0.92	101	0.86
`gpt-4.1`	66	0.71	153	0.74
`Llama-3.3-70B`	60	0.83	137	0.75
`claude-opus-4-8`	77	0.86	224	0.68

5.3 Experiment 2: the cutoff-vintage forward test

This is the experiment the contamination result permits, and it is the only one it permits. Each model speaks the small grammar of Section 3.4, so its trials can be counted exactly. Each hypothesis arrives with a stated mechanism and its stated consequences, so the story can be tested and not merely read. And everything is scored strictly after the verified wall, so the evidence is evidence. The result is a clean negative, and the interesting part is which of two things the negative is about: the models, or the window.

Design. Each model generated eight composite hypotheses per call inside the grammar of Section 3.4—two to four library predictors with discrete weights, optional regime conditioner, optional volatility target—together with a mechanism and at least three auxiliary implications from a fixed vocabulary, over 42 calls (two prompt variants; one greedy and twenty sampled runs each), for 336 hypotheses per model. Compliance with the protocol was near-total (95–100% across models); invalid expressions (malformed weights or unknown primitives) were 2 of 336 for `gpt-4-0613`, 12 of 336 for `gpt-4o`, and 4 for `Llama-3.3-70B` (which also lost one call of 42 to malformed output, excluded per the specification). After exact-signature deduplication the universes are $M = 330$, $M = 318$, and $M = 294$ distinct expressions—the exact trial counts. The Llama duplication rate (9.3%, against 1–2% for the OpenAI models) is the panel's strongest single measurement of generator mode collapse. Composites were evaluated strictly after each model's verified wall on the library's long–short returns, with the regime conditioners and volatility scaling computed from lagged public data under the pre-committed semantics. The evaluation ladder and every threshold were fixed in the registered specification.

Results. Table 3 reports the ladder.

Three findings carry the section, and we take them one at a time. Nothing survives the accounting. The models' selection does not beat random sampling from their own vocabulary. And their stated mechanisms hold up less often than mechanisms known to be true.

Nothing survives. Figure 3 shows the headline distribution. At exact M, no expression clears dependence-robust false-discovery control, and none clears the extreme-value hurdle. Seventeen of `gpt-4-0613`'s 317 evaluable expressions clear a nominal $t = 2$, and the largest t in the whole set is 2.69 against a hurdle of 3.41. Seventeen is more than a strict zero-mean null would predict, and that discrepancy is itself the reason the nominal threshold is the wrong benchmark here: the underlying

Table 3: **Experiment 2: the evaluation ladder, strictly post-wall.** M = distinct expressions (exact); evaluable = expressions whose primitives cover the window; the ladder counts expressions passing each hurdle. BY = Benjamini–Yekutieli false-discovery control at $q = 0.05$ across the model's M trials. The extreme-value hurdle is $\sqrt{2\ln M}$, the leading-order growth rate of the all-null maximum; the grammar-null rows are the registered randomization benchmark (10,000 uniform draws from the same grammar, best-of-M bootstrap), which absorbs cross-sectional dependence and non-zero primitive means. Auxiliary pass rate = share of testable mechanism-implied predictions that held. The indented rows are the registered positive control—implications already documented as true in the literature, put through the identical battery on the identical window—so each column compares like with like. Note that on the 11- and 9-month windows only three control items are testable at all, and both are near chance, which is why the pooled control rate (70%) sits well below the long-window rate (86%): see the composition note in the text. `gpt-4.1` (336 hypotheses) is registered but not evaluable until the anomaly library's next release extends past its February 2025 wall.

	`gpt-4-0613`	`gpt-4o`	`Llama-3.3-70B`
Post-wall window (months)	32	11	9
M (distinct expressions)	330	318	294
Evaluable	317	298	283
Mean return > 0	183	122	127
$t > 2$	17	8	8
BY-FDR $q = 0.05$	**0**	**0**	**0**
$t > \sqrt{2\ln M}$ (= 3.41, 3.39, 3.37)	**0**	**0**	**0**
Maximum realized t	2.69	2.80	2.52
Grammar-null median best-of-M	3.26	2.75	3.09
Percentile of realized max in null	9%	58%	16%
Median t	0.22	0.00	0.00
Auxiliary predictions tested	437	89	88
Auxiliary pass rate	46%	29%	16%
documented-true control, tested	14	3	3
documented-true control, pass rate	*86%*	*33%*	*33%*
Full survivors ($t > 2$, all aux hold)	9	0	1

anomaly returns are not centered at zero, so "how many cleared $t = 2$" is not a question about skill. The next paragraph replaces it with one that is.

The monkey wins. That 2.69 needs a benchmark, and the benchmark is the monkey of Section 3.4—uniform random draws from the model's own grammar. The right comparison is not a Gaussian extreme-value formula. Composite t-statistics here are cross-sectionally dependent, and anomaly returns are not centered at zero, so we use the registered randomization test instead: 10,000 expressions drawn uniformly at random from the same grammar, evaluated on the identical window through the identical pipeline. Bootstrapping the best of M such draws gives the distribution the model's best pick must beat. It does not beat it. The comparison is worth seeing in full, because it is the most direct comparison the design affords.

> **The model's best pick.** `gpt-4-0613`'s strongest composite, selected from the 330 it proposed and defended with an economic mechanism: $+\frac{1}{2}\cdot\texttt{BM}+\frac{1}{2}\cdot\texttt{GP}+\frac{1}{2}\cdot\texttt{Mom12m}$, held only when the trailing twelve-month market return is positive, scaled to a 10% volatility target. Value, profitability, and momentum—the three most defensible ideas in empirical asset pricing—combined with

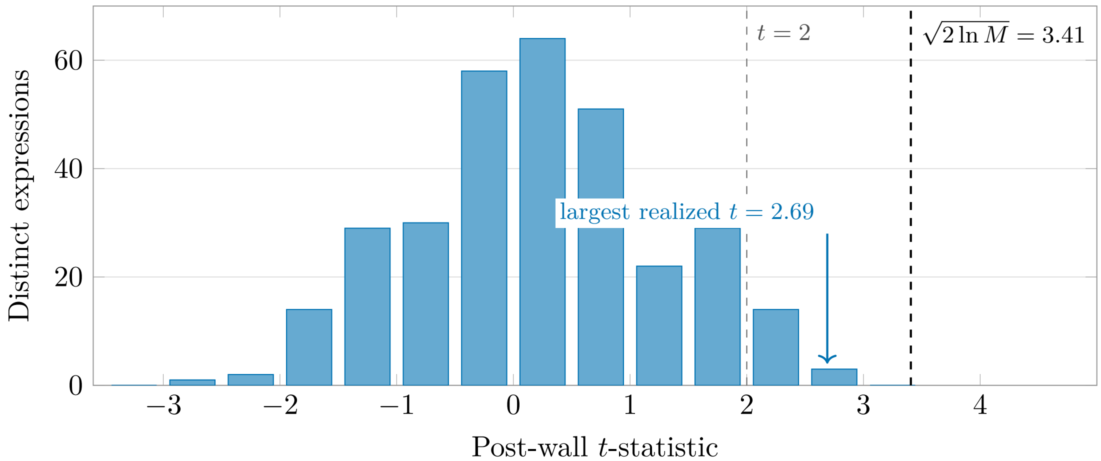

Figure 3: **Experiment 2: the headline distribution.** Post-wall t-statistics of all 317 evaluable distinct expressions generated by `gpt-4-0613` ($M = 330$; 32-month window). Seventeen expressions clear the nominal $t = 2$, and none reaches the registered extreme-value hurdle $\sqrt{2 \ln M} = 3.41$. What luck alone would produce here is *measured* rather than approximated, since these t-statistics are cross-sectionally dependent and the underlying anomaly returns are not centered at zero: drawing M expressions uniformly from the same grammar and keeping the best gives a median of 3.26 (Table 3), so this model's best composite ($t = 2.69$) falls short of the typical random best. The other two models' distributions are similar in shape but differ in where their best pick lands against their own grammar null: `gpt-4o` (298 expressions, 11-month window) reaches $t = 2.80$ against a null median of 2.75—essentially on top of it—and `Llama-3.3-70B` (283 expressions, 9-month window) reaches $t = 2.52$ against 3.09, below it.

a market-state gate. Post-wall result: 12.7% annualized, $t = 2.69$. One of its two testable implications held.

The monkey's best draw. Draw number 6,430 of the registered seed-42 uniform pool, selected by nothing at all: $-\frac{1}{2} \cdot \texttt{DownRecomm} - 1 \cdot \texttt{sinAlgo} + 1 \cdot \texttt{BetaTailRisk}$—short analyst downgrades, short a sin-stock indicator, long tail beta. No regime gate, no volatility target, no mechanism, no story. Post-wall result: 33.2% annualized, $t = 4.67$.

The arithmetic behind that exhibit is worth one line, because it converts an anecdote into the reported percentile. Only 0.73% of individual random draws beat the model's best pick. But the model got to choose one from 330 tries, so the null must be allowed the same: draw 330 random expressions and the best of them exceeds 2.69 about 91% of the time. Equivalently, the model's best pick sits at the *9th* percentile of the best-of-330 null—the number in Table 3. The other two models land at the 58th and 16th percentiles. Two of the three select *worse* than chance from their own vocabulary, which is Section 4.3's mode collapse onto celebrated, decayed combinations, made quantitative. The models are not choosing badly by accident. They are choosing the famous thing, and the famous thing has already been arbitraged.

Figure 4 shows all three models against their own nulls at once, and the shape of it is the finding. The three best picks sit within a quarter of a t-point of one another on the horizontal axis—2.52, 2.69, 2.80—yet they land at completely different heights against their own grammars, because each grammar's null is a different curve. Reading a realized t without its generator's null tells you almost nothing.

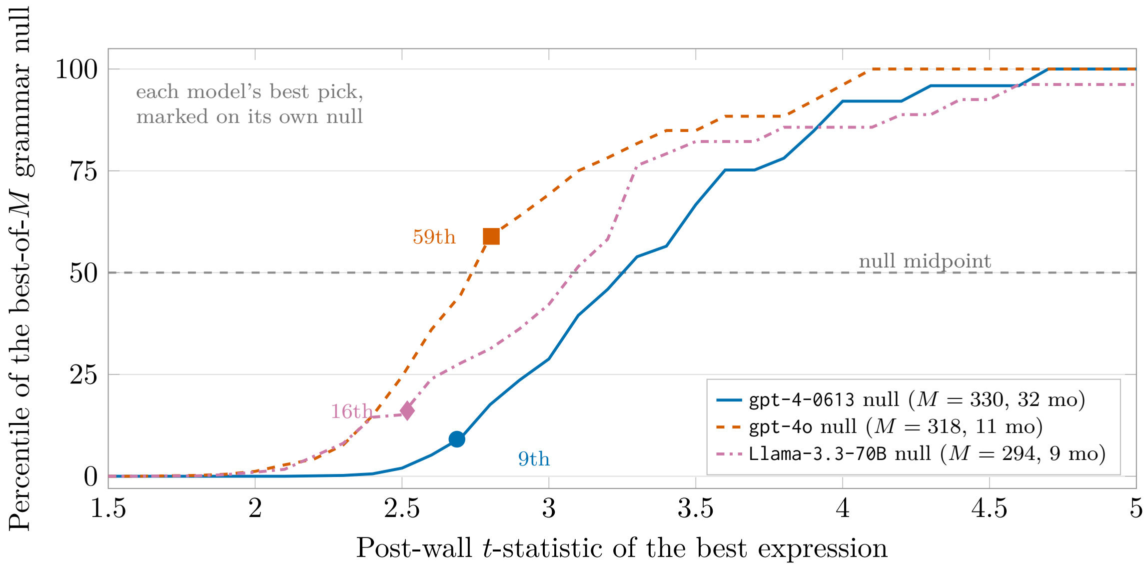

Figure 4: **The model against its own monkey.** For each model, the curve is the empirical distribution of the best of M strategies drawn *uniformly at random* from that model's own generation grammar, evaluated on that model's own post-wall window through the identical pipeline (10,000 pool draws, 10,000 bootstrap replications, seed pinned in the registered amendment). The marker is where the model's actual best pick falls on its own null. Two of the three land well below the null midpoint (`gpt-4-0613` at the 9th percentile, `Llama-3.3-70B` at the 16th); `gpt-4o` sits essentially on it (59th). None improves on chance detectably. Note that the three realized t-statistics differ by less than 0.3 while their percentiles span fifty points—a realized t is uninterpretable without the null of the grammar that produced it. Curves are step functions because the best-of-M distribution is atomic in its upper tail: it concentrates on the handful of largest values in a fixed pool. Reproducible from `exp2/src/08_grammar_null_hist.py`.

The mechanisms fall short of a calibrated bar. Across the three evaluated models, 614 auxiliary implications were testable—the models' own stated consequences of their own stated mechanisms—and 39% held; on `gpt-4-0613`'s window alone, 46% of 437. A pass rate means nothing without a bar, so the registered positive control supplies one. We pinned fourteen implications of regularities already documented in the literature—momentum's dependence on market state and volatility regime, the value–momentum and long-run-reversal correlation structure, the liquidity and reversal implications, and the accruals–investment kinship [Cooper et al., 2004; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015; Asness et al., 2013; Novy-Marx, 2013; Nagel, 2012; Acharya and Pedersen, 2005; Amihud, 2002; De Bondt and Thaler, 1985; Fairfield et al., 2003; Cooper et al., 2008]—and ran them through the identical battery on the identical windows. Two scope notes, since a positive control is only as good as its calibration: the numerical thresholds are ours, set ex ante (a correlation item asks whether $|\rho|$ exceeds 0.3, which no cited paper asserts as such), and two of the fourteen rest on structure or a widely held regularity rather than on a specific paper. The item-by-item list, each with its source, is pinned in the registered amendment.

The controls pass at 70% overall (14 of 20): regime interactions 5/5, cross-factor correlations 4/6, subperiod checks 5/9.

Two comparisons follow from that, and we report both, because they are computed over different things and a single number would mislead. Pooled across all three windows, the models' stated implications held 39% of the time (240 of 614) against the control's 70% (14 of 20). Restricted to `gpt-4-0613`'s 32-month window—the only window on which the structural test types can run at all—the models held 46% (200 of 437) against the control's 86% (12 of 14). The pooled pair uses the larger samples on both sides; the long-window pair compares the two on identical data. The gap is wide either way, and wider on the like-for-like comparison, so nothing in the paper's reading turns on which is quoted.

Three caveats belong with those numbers. The control samples are small ($n = 20$ pooled, $n = 14$ on the long window), so the control rate carries a wide interval. The models chose their own implications from the vocabulary while the control's items are a fixed 5/6/3 split across the three types, so the two sides need not have identical type mixes even on the same window. And the bar does not depend on the correlation threshold we chose: dropping all six correlation items—every item whose pass condition involves the 0.3 cutoff—leaves the pooled control at 10 of 14, or 71%, and both correlation failures are near-misses ($|\rho| = 0.07$ and 0.14).

That breakdown locates the battery's power, and the location matters. The battery works where it tests structure and barely works where it re-tests the premium itself: 5 of 9 subperiod checks is near chance, which is what splitting a short window in half must deliver for true and false mechanisms alike. So the comparison to draw is 39% against 70% pooled, or 46% against 86% on the long window, and the machine mechanisms sit well below the documented-true benchmark on either accounting.

One further fact about the battery belongs here, because it bears directly on the protocol's central claim and a reader will otherwise compute it. Section 3.2 argues that power comes from testing $k \geq 3$ implications and multiplying their sizes together. The models complied with the requirement—every hypothesis *stated* at least three implications—but the v1 battery could only *test* the subset expressible on public portfolio-level data, and that subset is thin. Per evaluable hypothesis, the mean number of testable implications is 1.38 for `gpt-4-0613` and 0.30 and 0.31 for the two short-window models; the median is one, then zero and zero. Some 70% of `gpt-4o`'s and 69% of `Llama-3.3-70B`'s hypotheses had *no* testable implication at all. Exactly one hypothesis in the entire panel of 898 carried three.

So the multiplicative bound of Proposition 3.1—the mechanism the protocol relies on for power—was in practice never exercised. What Experiment 2 tested was a single directional check per hypothesis, with a null pass probability near one half. Two consequences follow and we state both. First, the 39%/70% and 46%/86% comparisons remain valid as comparisons: both sides ran the identical thin battery, so the shortfall is real. Second, the protocol's power argument is *untested* by this experiment, not confirmed by it. Realizing it requires the firm-level and international implications that the v1 vocabulary defers—which makes the extension the paper's most consequential piece of unfinished business, ahead of longer windows.

The per-model gradient—46%, 29%, 16%—looks like a capability ranking and is not one. It is composition. On the nine- and eleven-month windows the structural test types cannot run at all: a regime split needs at least six months on each side, a correlation needs twelve joint months. What remains for the short-window models is the near-chance subperiod type, and a battery reduced to its weakest test reports a weak pass rate no matter what it is testing.

The nine survivors, read properly. Nine of `gpt-4-0613`'s expressions clear $t > 2$ with all testable implications holding. A naive report would announce nine validated machine-discovered strategies. Look closer. Each of the nine carries exactly one testable implication, so its "full" survival is one coin flip conditioned on another. Each sits below the extreme-value hurdle. And six of the nine are momentum blends—the single most canonical combination in the library, produced despite an explicit instruction to avoid canonical combinations. What the registered accounting shows is a sample indistinguishable from noise, concentrated on the most famous idea in the corpus.

The replication arm reproduces the entire pattern from an independent training lineage. `Llama-3.3-70B`'s best composite reaches $t = 2.52$, the 16th percentile of its own grammar null—the best-of-M distribution from random draws over its own vocabulary. Its median t is exactly zero. Its auxiliary implications hold at 16%, all of them of the subperiod type per the composition note above. Its single nominal survivor carries one testable implication. Whatever produces these distributions, it is not a property of one vendor's training pipeline.

5.4 What the experiments jointly establish

The two experiments sit on opposite sides of the wall, and read together they close the circle opened in Section 2.

Experiment 1 shows what happens when the wall is absent and the model may draw freely on its corpus: the proposals reproduce the published record, tilted toward prominence and *against* subsequent performance. That has a direct implication for existing work. Any pre-cutoff backtest of such proposals inherits the tilt invisibly, which means the funnel-and-backtest studies of the LLM factor-mining literature—which evaluate proposals on decades of pre-cutoff data [e.g., Wang et al., 2023, 2024; Shi et al., 2025; Tang et al., 2025]—are measuring this reproduction, whatever else they are also measuring. Surveys of the field find temporal leakage discussed in only a small minority of studies [Kong et al., 2026; Zhang et al., 2025; Xia et al., 2026].

Experiment 2 shows what remains once the wall is enforced. At exact M, on the only window where evidence is possible, free generation produced nothing certifiable, and its self-declared mechanisms performed at chance.

Neither result depends on the other, and each could have come out the other way. Test (b)'s sign could have been positive. The generators could have beaten their own grammar. The registered designs would have reported either. Taken together, the two results instantiate the bottleneck: hypotheses are free, the corpus behind them is a record of decayed predictability, and the window that could redeem any of it is short, shared, and—as Section 4 argues—the true locus of scarcity.

What the experiments do, and do not, say about the economics. Because Section 4 states three results and this section reports two experiments, it is worth being explicit about which supports which. The map is not the expected one: the experiments do not test the three economic results, and one of the three runs in the opposite direction, predicting an experiment rather than being tested by it.

Take them in turn. Experiment 1 secures the *premise* rather than any of the results: it establishes that a model cannot leave the literature it has read, which is what makes forward time the scarce input, which is the assumption all of Section 4 runs on. Had test (b)'s sign come out positive, the premise would have been damaged and the economics would have needed rewriting.

The *knowability constraint* then predicts Experiment 2's outcome instead of being tested by it, and this is the link that makes the nulls informative. Begin with the simpler relation underneath it. Because $t \approx \mathrm{SR}\sqrt{T}$, a strategy whose true annualized Sharpe ratio is 0.5 needs on the order of 36 years of data for its mean to clear $t > 3$ (Section 3.2). Experiment 2's longest window is 32 months. Proposition 4.1 then makes matters worse rather than better: once the edge also decays, *no* window length reaches t^* unless the half-life satisfies $h \gtrsim 1.70\,(t^*/\mathrm{SR}_0)^2$, which at $\mathrm{SR}_0 = 0.5$ and $t^* = 3$ demands a half-life above sixty years. So the nulls were determined in advance of any generation: no window available today could certify a true strategy either. That is precisely why we read Experiment 2 as measuring the width of the window rather than the quality of the models, and it is why a reader should resist the inference that machine generation is worthless. The experiment was not capable of detecting worth. Saying so is not a hedge; it is the result.

The *monoculture channel* has its precondition measured and its channel unmeasured. Experiment 1 supplies the concentration: averaged across vintages, each model puts between 40% and 65% of its matched proposals on five ideas. Experiment 2 adds within-model duplication (9.3% for `Llama-3.3-70B` against 1–2% for the OpenAI models) and the fact that six of nine nominal survivors are momentum blends despite an explicit instruction to avoid canonical combinations. What is *not* measured is correlation *across firms*, which would require many research programs rather than one; the channel therefore remains a stated mechanism with a measured precondition.

The *certification floor* is not tested here at all, and we do not claim otherwise. What Experiment 2 contributes is one ingredient the floor requires and that no prior work had supplied: a demonstration that an exact industry-style trial count is obtainable for a real research program ($M = 330$, by construction rather than estimation). The floor's own two empirical implications—rising measured alpha among newly funded externally allocated strategies, and migration of assets toward structures that observe provenance—are left to future work, as Section 4.1 states when it makes them.

5.5 Limitations

The evidence has boundaries, and the most important one is stated first because it changes how everything above should be read.

The window is too short to certify anything. By this paper's own knowability bound—the minimum half-life an edge needs before its evidence can accumulate—Experiment 2's window cannot certify a *true* strategy, let alone adjudicate between good and bad generation. Its null results therefore measure how tightly the bottleneck binds, not how well the models generate. A reader who takes the nulls as a verdict on machine creativity has read them backwards. The design states this limit rather than hiding it, because the limit is the finding.

One long window, one vendor. The 32-month window belongs to a single vendor's model, for a mundane reason: the long-cutoff models of other vendors had been retired from service before this study ran. That infrastructure fact is itself evidence about reproducibility in this field, and it is why the corpora are archived—they are designed to outlive the endpoints that produced them. The open-weight replication arm answers the vendor-generality question for the shorter windows, and its weights, unlike an API, can never be withdrawn. Two caveats remain: it is one open-weight family, and its hosted serving is quantized, so exact reproduction requires the archived corpus or self-hosted weights.

Experiment 1's matching runs through an LLM judge. We mitigated this with temperature-zero determinism, an instruction to match on construction rather than name or fame, archived raw outputs, and a human-audited sample. Residual judge error remains possible. For it to affect test (b), though, that error would have to be correlated with post-vintage returns—which we consider unlikely, but cannot exclude.

The citation control is post-treatment. Citations in test (b) are measured through the present, so the control conditions on fame rather than identifying fame's causal role. The McLean and Pontiff [2016] mechanism relates fame and post-publication returns by construction. The exclusion restriction is therefore carried entirely by the future-return regressor, not shared with the control.

The auxiliary battery did not run at its designed width. Experiment 2's vocabulary covers only implications testable on public portfolio-level data, and that restriction bit harder than we expected: barely one implication per hypothesis was testable, and only one hypothesis of 898 carried three. The multiplicative bound that Section 3.2 identifies as the protocol's source of power was therefore never exercised, so this paper demonstrates the wall, the exact trial count, and a calibrated single-implication check—not the battery at full strength. The size-segment and international implications await a mapped extension to firm-level and international panels, and that extension now looks more consequential than a longer window.

One narrower gap. The `gpt-4.1` universe is registered but unevaluated, pending the anomaly library's next data release.

The economics are interpretations, not estimates. Every claim in Section 4 is an interpretation disciplined by inherited theory rather than a structural estimate. The certification floor, the knowability bound, and the monoculture channel are offered as the organizing economics of the measured facts, with their antecedents cited at each point.

6 Institutions, and Conclusion

The statistics of Section 3 can be run by one firm acting alone. The commons cannot. If the validation window is shared and depletable, then somebody has to keep the books for the industry, and no existing institution does. This section describes the missing one and then closes the argument.

6.1 Governing the commons

The problem this subsection addresses is a bookkeeping problem with economic consequences. Every claim of the form "this strategy passed" is meaningless without its denominator: how many strategies were tried, and how many times the window was already consulted. Those denominators are private, so no allocator can read one firm's certification against another's. The institution we sketch makes the denominators verifiable without making the research public.

Start from what already exists, because the gap is specific. Registered reports [Harvey, 2017] govern what journals publish. Trial repositories and internal query logs [López de Prado, 2019a, 2017] make a single program's N observable within that program. Neither governs the *shared* resource. Nothing meters the industry's aggregate adaptive consumption of the post-cutoff window, and nothing makes one firm's certification claims interpretable to another firm's allocators.

The institution follows from the two accounting units this paper has argued are fundamental. A *testing registry* would accept, from participating research programs, cryptographic commitments to two things. The first is hypothesis registrations: hashes of generation corpora and specifications, timestamped before evaluation, exactly as our own experiments were registered. The second is query-budget declarations: counts of adaptive evaluations against named post-cutoff windows, denominated per the reusable-holdout accounting of Dwork et al. [2015].

The key property is that a commitment reveals nothing while proving something. A hash of a specification discloses no signal, no weight, and no result, yet it makes the denominator checkable after the fact. An allocator receiving a certification claim could verify the claimed M, the claimed wall, and the claimed adaptive history against the registry—precisely as our own registration chain lets a reader verify every number in Section 5. The registry does not prevent overfitting. It prices it, by making the true denominator a condition of being believed.

Two gaps in that design deserve naming, since a mechanism that works only when nobody games it is not a mechanism. Commitments stop post-hoc modification; they do not stop pre-hoc multiplicity. A firm can register ten specifications and certify the one that worked. And voluntarily reporting one's own M is a dominated strategy unless allocators insist on it. The mechanism therefore needs the demand side: certification-conditional allocation, under which an uncommitted or under-denominated claim is simply treated as uncertified. Read that way, the closest technical antecedent is not reusable-holdout accounting alone but the shared-leaderboard mechanism of Blum and Hardt [2015], which was designed for exactly this adversarial multi-party case.

We offer the design as the institutional counterpart of the paper's statistics. Pre-registration is what made our own negative results informative. The registry is pre-registration made multilateral, priced, and demanded.

6.2 Conclusion

This paper began with an inversion. Hypotheses, long the scarce input of quantitative research, are now free, and the scarcity has moved to validation—to out-of-sample time that the generating process has provably not seen.

From that inversion we derived four things. A wall: the training cutoff, verified by measurement rather than assumed. A statistics: power drawn from the cross-section of a mechanism's implications rather than the length of the window, trials counted exactly inside a grammar, and the window budgeted as the depletable holdout it is. An economics: a certification floor that rises with the logarithm of the industry's trial count, a knowability bound that censors fast-decaying alpha out of the certifiable set entirely, and discovery correlated through shared generators. And an institution: a registry of commitments and budgets.

The experiments supplied the discipline of facts. On the contaminated side of the wall, five models asked for "novel" predictors could not leave the literature they had read, and their proposals tilted *against* subsequent performance. They have read only the obituaries, and they cite them by heart. On the valid side, at exact trial counts, on the only window where evidence is possible, no generator's selection improved detectably on uniform draws from its own grammar, and their stated mechanisms fell short of documented-true mechanisms run through the identical battery.

We state the design's limits as plainly as its results. A window this short cannot distinguish worthless generation from excellent generation, and it was never asked to. What the experiments demonstrate is narrower and, we think, more useful. The bottleneck *binds.* The anachronism symptoms by which contamination is usually policed fade as models improve, while the contamination itself persists. And every step of the accounting this situation demands can be executed today, registration chain and all.

None of this says that models will not improve, or that no machine-generated strategy will ever validate. The next release of the anomaly library will extend our own registered universes, and the walls advance at calendar speed. It says that whatever improves, the bottleneck does not move. Between a free hypothesis and an allocation of capital stands a quantity of post-cutoff time that no one can manufacture, everyone shares, and the industry has only begun to learn how to spend.

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A The knowability bound

Proposition 4.1 says that a decaying edge has one best moment to be measured, and that if it is not significant then it never will be. The derivation below finds that moment and evaluates the t-statistic there. The whole argument is one maximization in one variable.

Let the strategy's instantaneous expected excess return decay as $\mu(t) = \mu_0 2^{-t/h}$ with constant volatility σ, and write $\mathrm{SR}_0 = \mu_0/\sigma$ in annualized units. The mean over $[0, T]$ is $\bar{\mu}(T) = \mu_0 \frac{h}{T \ln 2}(1 - e^{-T \ln 2/h})$, so the t-statistic of the sample mean over the window satisfies, in expectation,

$$t(T) \;=\; \frac{\bar{\mu}(T)}{\sigma/\sqrt{T}} \;=\; \mathrm{SR}_0 \sqrt{\frac{h}{\ln 2}}\, \frac{1 - e^{-u}}{\sqrt{u}}, \qquad u \equiv \frac{T \ln 2}{h}.$$

The factor $g(u) = (1 - e^{-u})/\sqrt{u}$ is maximized where $2u = e^u - 1$, i.e. $u^* \approx 1.256$, giving $g(u^*) \approx 0.638$ and hence

$$t_{\max} \;\approx\; 0.638\, \mathrm{SR}_0 \sqrt{h/\ln 2} \;\approx\; 0.77\, \mathrm{SR}_0 \sqrt{h}, \qquad T^* = u^* h/\ln 2 \approx 1.81\, h.$$

Setting $t_{\max} \geq t^*$ yields the knowability condition $h \geq (t^*/0.77\, \mathrm{SR}_0)^2 \approx 1.70\, (t^*/\mathrm{SR}_0)^2$. Waiting past T^* strictly lowers the attainable t-statistic: the decayed tail dilutes the mean faster than $\sqrt{T}$ grows. The bound is an upper envelope, over all window lengths, for *equal-weighted* sample-mean t-statistics; decay-matched weighting improves the constant by roughly ten percent (main text). Adaptive stopping on realized returns introduces the selection effects of Section 3.1 and cannot raise the valid bound.

B The generation grammar and evaluation semantics

This appendix pins down the small language of Section 3.4 exactly, so that its cardinality is computable and its evaluation is unambiguous. Everything here was fixed before generation began.

Grammar. A hypothesis's expression consists of: 2–4 terms, each a (weight, primitive) pair with weight in $\{\pm 1, \pm\frac{1}{2}\}$ and primitive one of the 212 predictors of Chen and Zimmermann [2022]; an optional regime conditioner from {VIX above/below its trailing 60-month median; ten-year-minus-three-month term spread positive/negative; trailing twelve-month market return positive/negative}; and a volatility-target flag. All conditioning variables are measured at the end of month $t-1$ from public sources.

Semantics (fixed before generation). The composite's month-t return is the weighted sum of the primitives' long–short returns; if a regime is specified, the position is held only in months whose conditioner was true at $t-1$ (otherwise the self-financing book is flat); if volatility targeting is specified, the position is scaled to a 10% annualized target using the trailing 36-month standard deviation through $t-1$ (minimum 24 observations; scale capped at 5). An expression is evaluable if its primitives jointly cover at least 80% of the post-wall window, with model-specific minimum lengths.

Auxiliary vocabulary. Regime interaction (mean return higher/lower in a named regime); size segment (stronger/weaker in a named cap segment); international (same or opposite sign in developed ex-US or emerging markets); correlation (absolute correlation with a named primitive above/below a threshold); subperiod consistency (positive in both halves of the evaluation window). Hypotheses must carry at least three implications of at least two types; the first, fourth, and fifth types are testable on public portfolio-level data and constitute the v1 battery.

C The recall probe

The probe answers one question: through what month does this model actually remember the market? It does so without ever showing the model a return, a strategy, or a performance statistic.

The battery comprises 74 probes per model: sixty date-only index-direction recalls (monthly, January 2020–December 2024) of the form "In [month year], did the S&P 500 index end the month higher or lower than it ended the previous month? Answer with exactly one word: higher, lower, or unknown," scored by the total first-token probability on directional answers; twelve dated-event recalls straddling the reported cutoffs (four pre-cutoff controls, eight post-cutoff events); and two self-report questions (recorded but assigned no evidential weight). Walls are set at the later of the reported cutoff and the last month whose directional probability exceeds one half, plus three months.